AI 모델 성능 개선 결과

아직 남은 것

- 관측 예측 하나 그래프로
- 대곡교 궁내교 수위 같이 그려서 비교

- 3시간 예측 R2 한번 확인
- 23년, 24년, 25년은 시점을 좀 앞으로 땡겨

결론

- 관심 강우+MAE 가 가장 안정적
- 20년 이후 학습 모델: 최신성우수, 22년 이벤트 음수

72시간 예측

72시간 예측 후 뒷부분 날리고 앞 36시간만 예측

* model_sungnam_(gn/dg)72

72시간 예측

- 음수 R^2 다수
- MAE/RMSE 모두 가장 큼

기본 모델 대비 평균 R2 : 궁내교 ▼ 저하 (- 62.3%) , 대곡교 ▼ 저하 (- 33.9%)

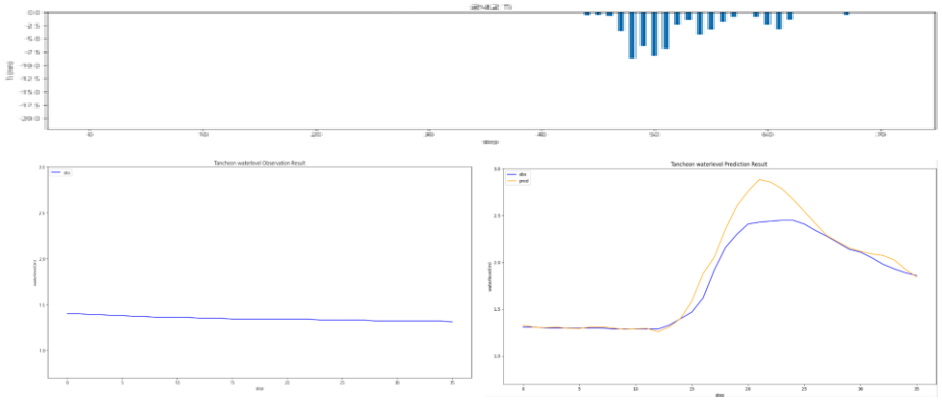
	궁내교				대곡교			
Test data	**r2**	**mse**	**mae**	**rmse**	**r2**	**mse**	**mae**	**rmse**
20.08.02	0.8759	0.0262	0.0960	0.1619	0.9934	0.0039	0.0495	0.0631
22.07.12	-1.9263	0.1161	0.3101	0.3407	-0.4222	0.1991	0.4156	0.4462
23.07.11	0.7968	0.0568	0.1515	0.2384	0.7761	0.2037	0.3916	0.4514
24.07.23	0.6043	0.0320	0.1075	0.1789	0.2256	0.0368	0.1646	0.1918
25.07.16	0.4520	0.0549	0.1735	0.2344	0.8822	0.0543	0.1855	0.2331
25.08.14	0.8114	0.0132	0.0951	0.1153	0.7106	0.1004	0.2621	0.3169

궁내교

* model_sungnam_(gn/dg)72

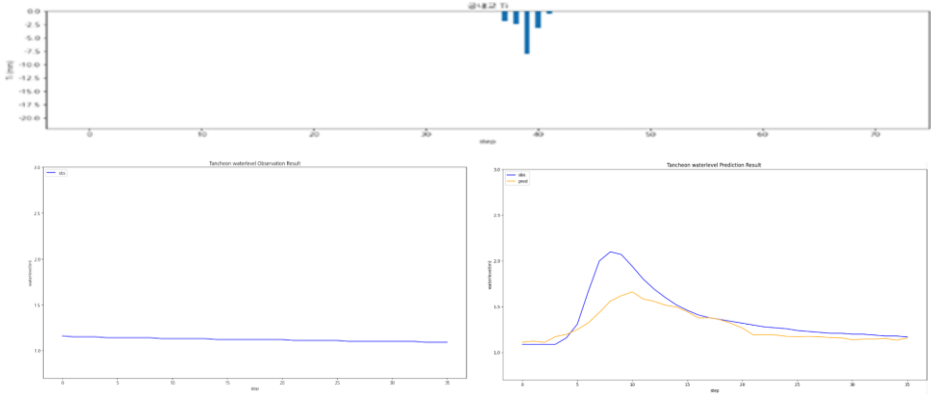
testdata_ 200802

R2 : 0.8759



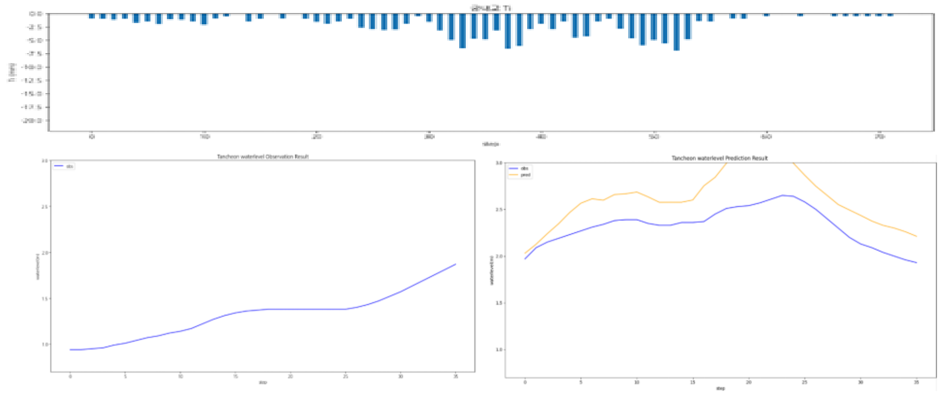
testdata_ 240723

R2 : 0.6043



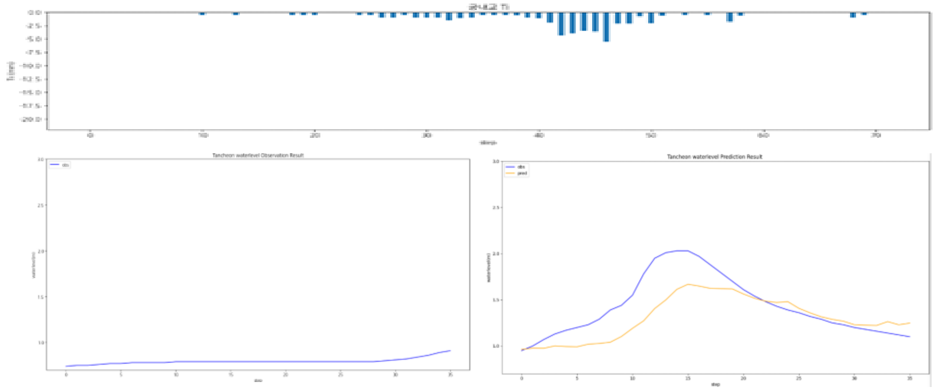
testdata_ 220712

R2 : -1.9263



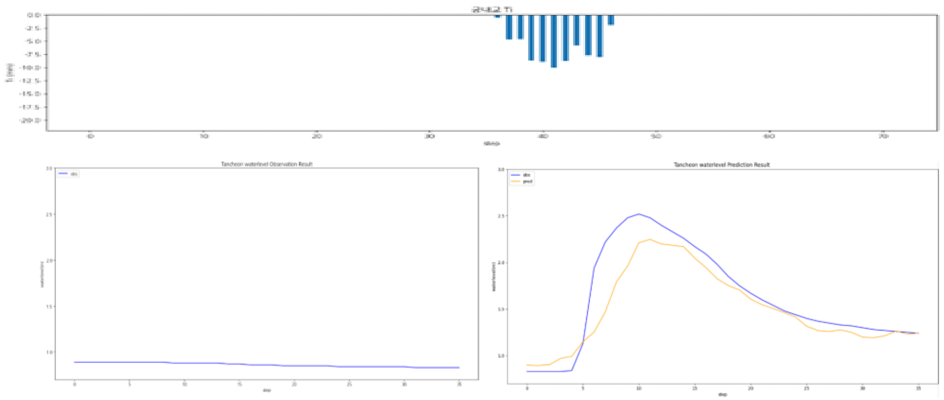
testdata_ 250716

R2 : 0.4520



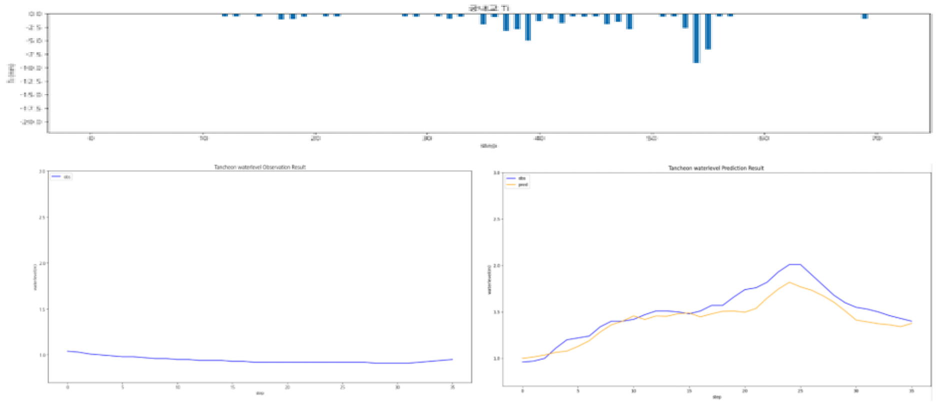
testdata_230711

R2 : 0.7968



testdata_250814

R2 : 0.8114

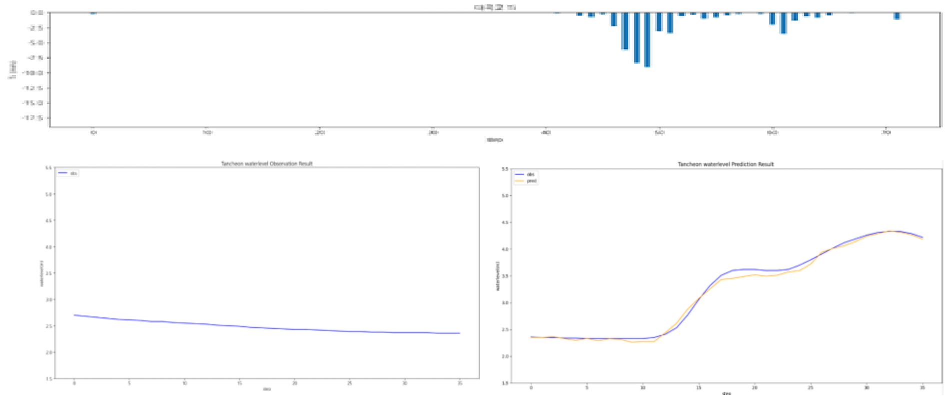


대곡교

* model_sungnam_(gn/dg)72

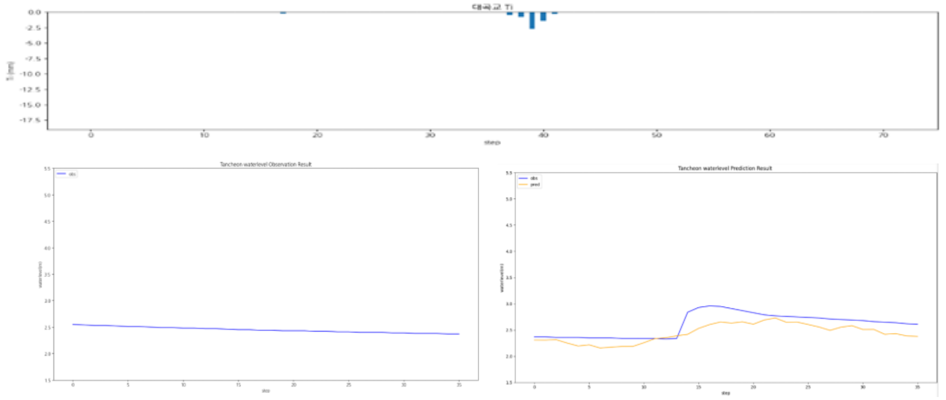
testdata_ 200802

R2 : 0.9934



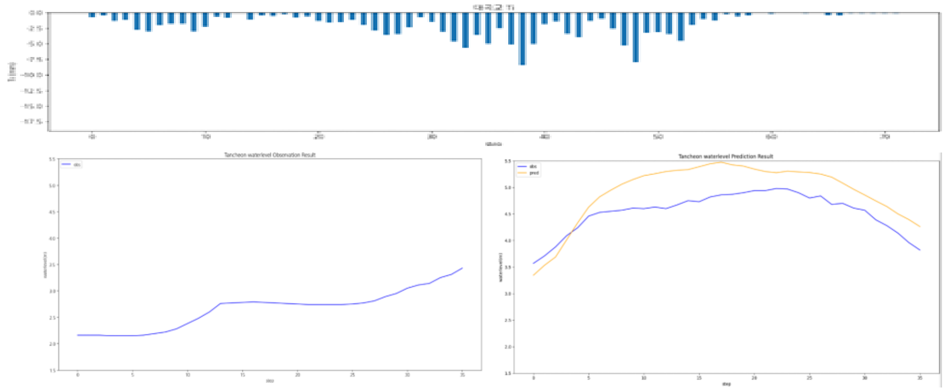
testdata_ 240723

R2 : 0.2256



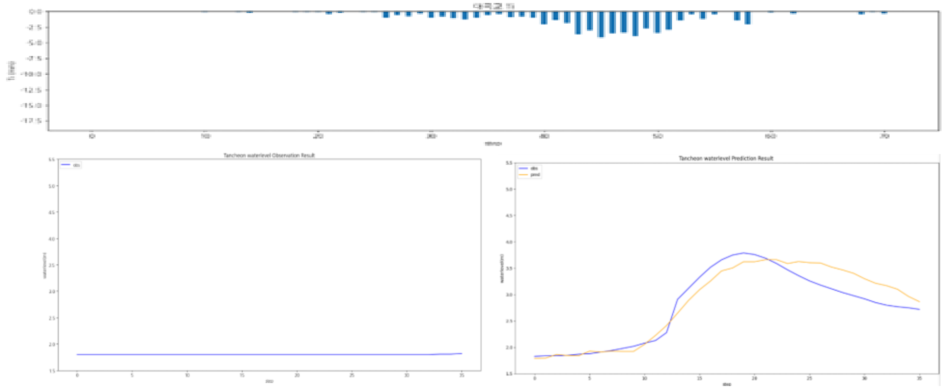
testdata_ 220712

R2 : -0.4222



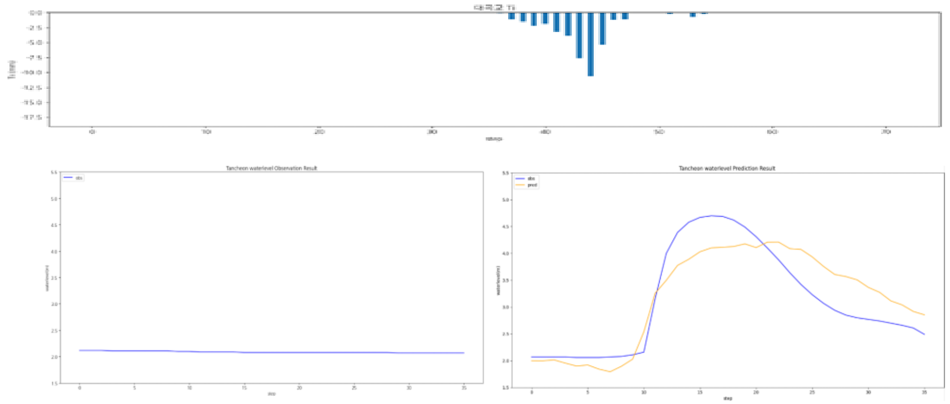
testdata_ 250716

R2 : 0.8822



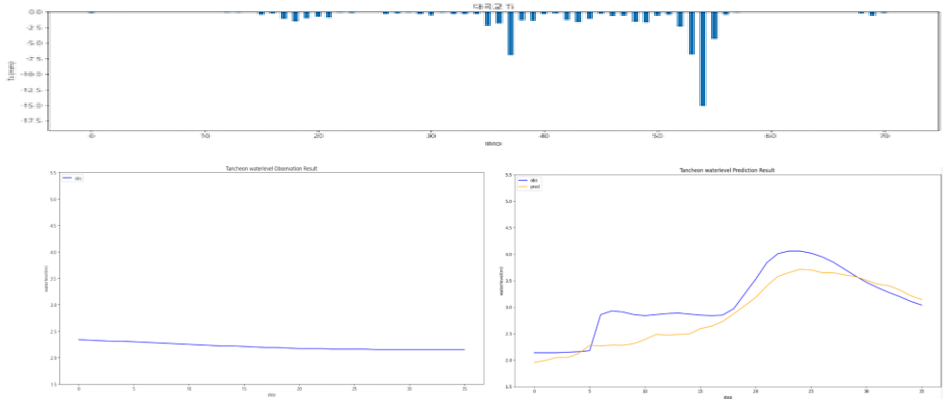
testdata_230711

R2 : 0.7761



testdata_250814

R2 : 0.7106



관심수위 + MAE

학습데이터

- 관심 수위 : 궁내교 (2) 대곡교 (3.8)
- smoothing : 궁내교 (1) 대곡교 (2)
- 그래프 스케일 고정 : 궁내교 (0.7 - 3) 대곡교 (1.5 – 5.5)

- learning_rate = 0.0005
- Loss_function = MAE
- Seq_length = 36 (6시간)
- Num_of_layer = 18

홍수통제소 API 호출 → MIET 활용 강우 이벤트 추출 → 관심 수위 이상 추출

time	서울시 (대곡교)	성남시 (성남북초교)	광주시 (남한산초교)	성남시 (대장동)	성남시 (구미초교)	성남시 (한국학중앙연구원)	성남시 (궁내교)_WL	성남시 (궁내교)_Q	서울시 (대곡교)_WL	서울시 (대곡교)_Q	대곡교_Ti	궁내교_Ti
2014-07-24 15:30:00	0.0	0.0	0.0	0.0	0.0	0.0	1.12	5.11	2.17	40.56	0.0	0.0

학습 , 검증 : 총 67개 中 61개 데이터 파일 사용

테스트 : 학습 데이터에서 제외한 6개 데이터 파일 (4개 이벤트)

+ 학습에서 검증에 사용된 2개 이벤트 사용

- 학습 : 2014 ~ 20230629 (43개)
- 검증 : 20230713 ~ 250813 (12개)
- 테스트 : 2550916 ~ 251006 (6개)

* model_(gn/dg)_wl18_3_2

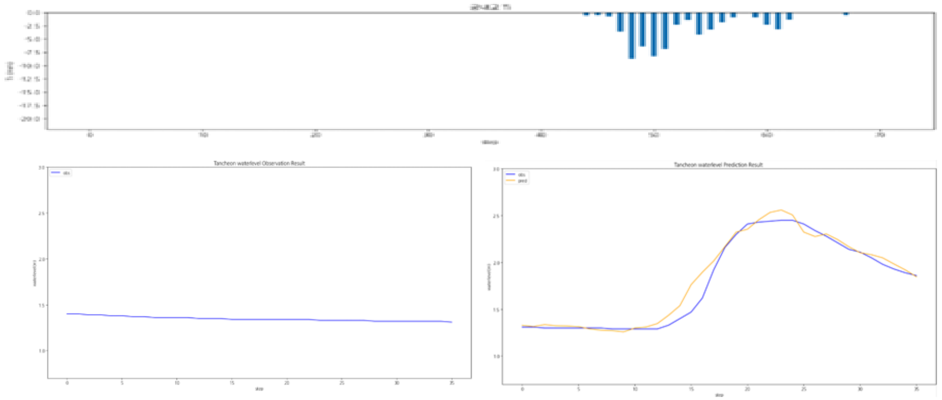
관심수위 + MAE

- 피크 타이밍 & 크기 가장 정확
- 기존 모든 실험 대비 가장 안정적
- 이전 모델 대비 피크값 추적력 , 고수위 예측 정확도 향상

	궁내교				대곡교			
Test data	**r2**	**mse**	**mae**	**rmse**	**r2**	**mse**	**mae**	**rmse**
20.08.02	0.9656	0.0072	0.0558	0.0852	0.8748	0.0763	0.1754	0.2763
22.07.12	0.4201	0.0230	0.1139	0.1516	0.5503	0.0629	0.2047	0.2509
23.07.11	0.8622	0.0385	0.1486	0.1963	0.8659	0.1219	0.2258	0.3492
24.07.23	0.7308	0.0217	0.0962	0.1475	0.6749	0.0154	0.0860	0.1244
25.07.16	0.7947	0.0205	0.1047	0.1435	0.9470	0.0244	0.1098	0.1563
25.08.14	0.5087	0.0346	0.1446	0.1861	0.8781	0.0423	0.1589	0.2056

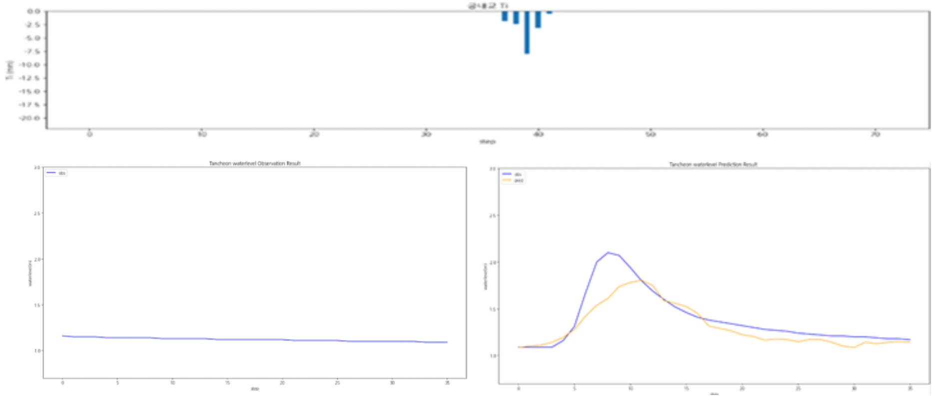
궁내교

* model_(gn/dg)_wl18_3_2



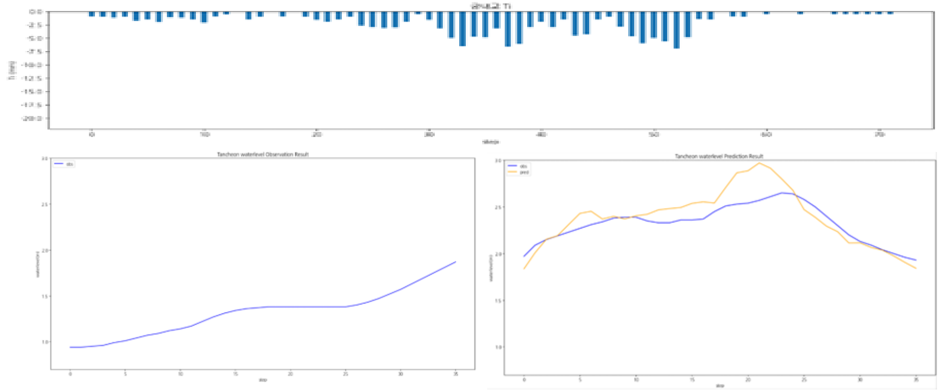
testdata_ 200802

R2 : 0.9656



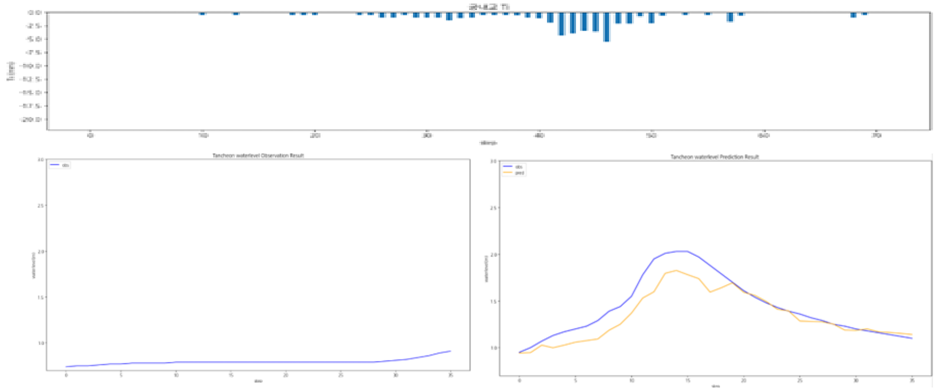
testdata_ 240723

R2 : 0.7308



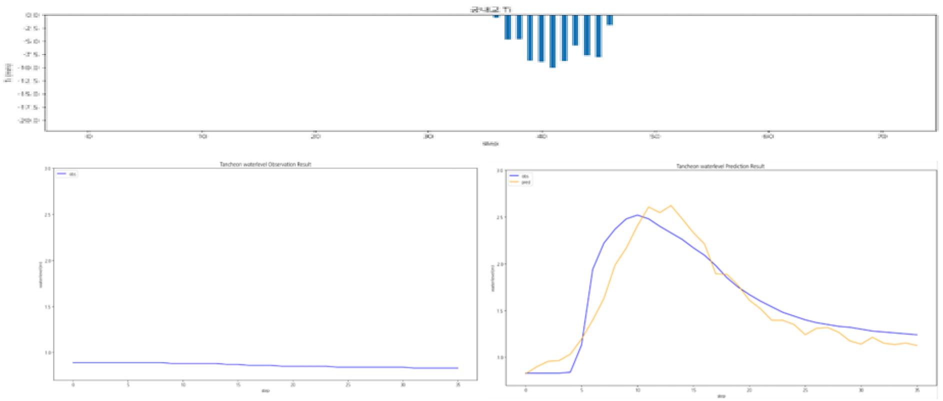
testdata_ 220712

R2 : 0.4201



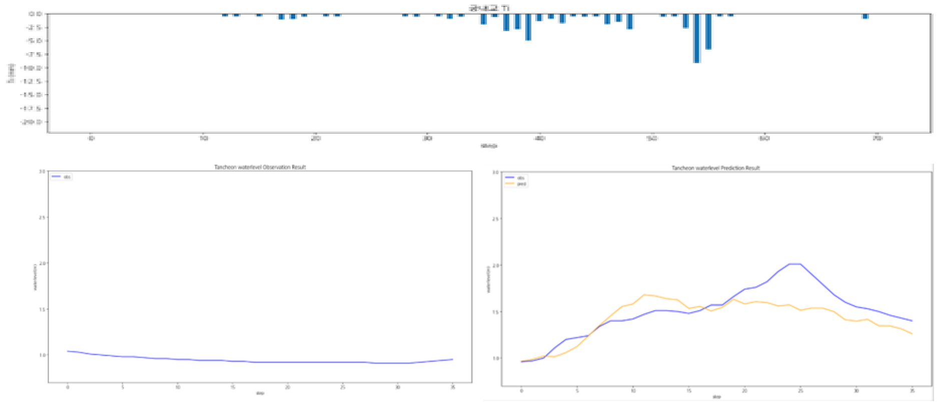
testdata_ 250716

R2 : 0.7947



testdata_230711

R2 : 0.8622



testdata_250814

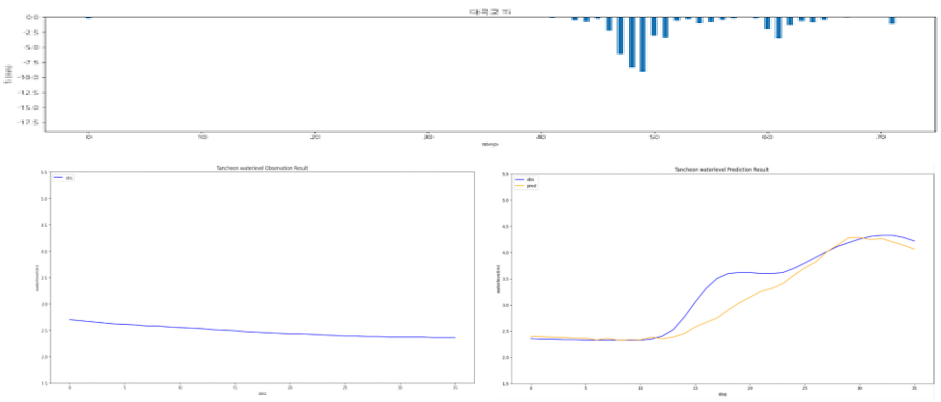
R2 : 0.5087

대곡교

* model_(gn/dg)_wl18_3_2

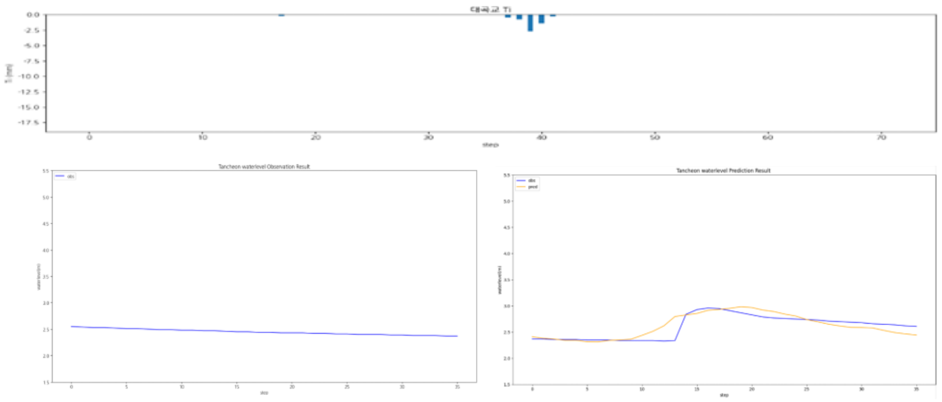
testdata_ 200802

R2 : 0.8748



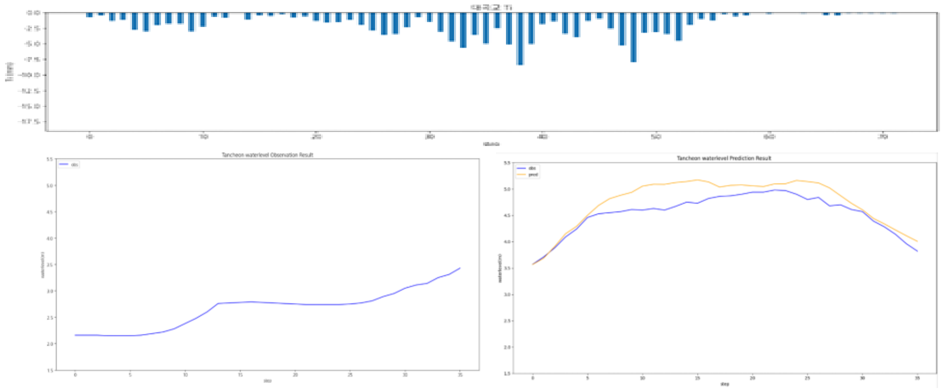
testdata_ 240723

R2 : 0.6749



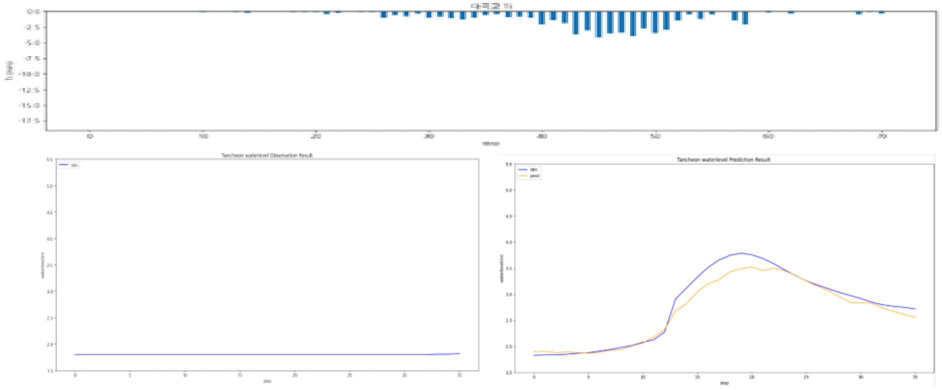
testdata_ 220712

R2 : 0.5503



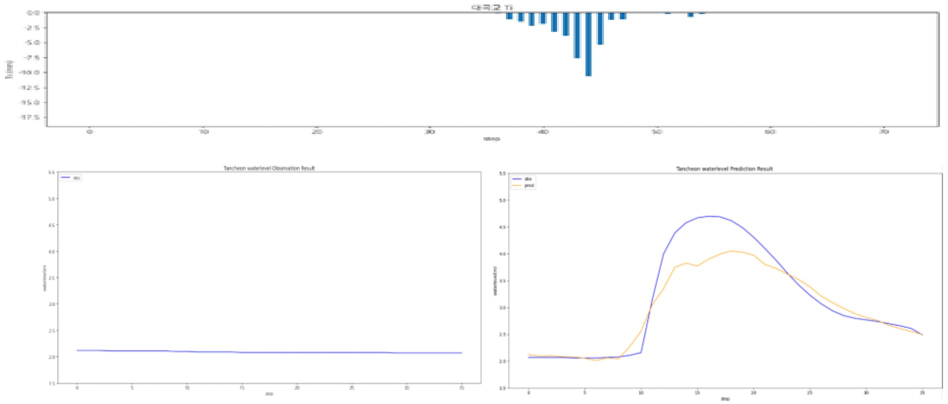
testdata_ 250716

R2 : 0.9470



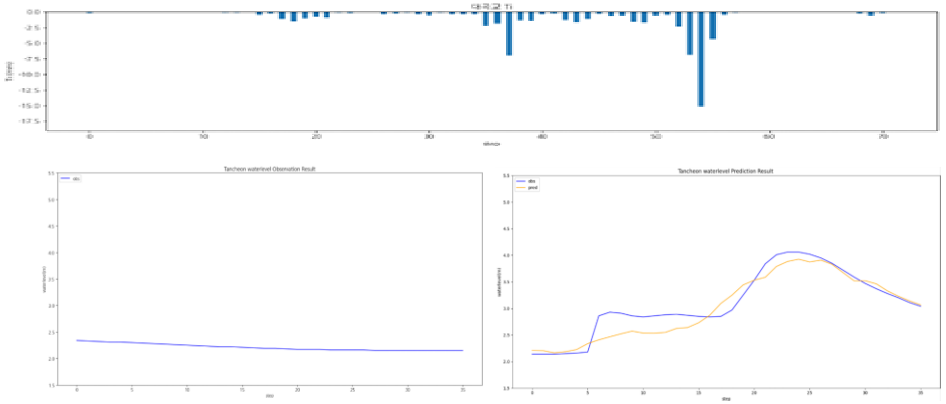
testdata_230711

R2 : 0.8659



testdata_250814

R2 : 0.8781



관심수위 + MAE
1시간 단위 예측

관심수위 + MAE 1시간 단위 예측

- 궁내교 : Lead time sensitivity 낮음 (시간 이동에 강건)
- 대곡교 : 궁내교보다 Lead-time sensitivity 큼 (특히 후행 1시간에서 성능 급락)
- peak 위치가 ±1h shift되어도 R^2 비슷하게 유지

		궁내교				대곡교			
Test data	**time**	**r2**	**mse**	**mae**	**rmse**	**r2**	**mse**	**mae**	**rmse**
20.08.02	**01**	0.9627	0.0085	0.0656	0.0656	0.8421	0.0716	0.1783	0.2676
	02	0.9656	0.0072	0.0558	0.0852	0.8748	0.0763	0.1754	0.2763
	03	0.8581	0.0236	0.1028	0.1538	0.8667	0.0633	0.2096	0.2517
22.07.12	**15**	0.8698	0.0108	0.0845	0.1041	0.9276	0.0271	0.1433	0.1648
	16	0.4201	0.0230	0.1139	0.1516	0.5503	0.0629	0.2047	0.2509
	17	0.7777	0.0148	0.0844	0.1218	0.3852	0.1563	0.3768	0.3953
23.07.11	**06**	0.8966	0.0368	0.1150	0.1920	0.8208	0.1877	0.2735	0.4333
	07	0.8622	0.0385	0.1486	0.1963	0.8659	0.1219	0.2258	0.3492
	08	0.8323	0.0351	0.1005	0.1874	0.9018	0.0779	0.2423	0.2791

* model_(gn/dg)_wl18_3_2

관심수위 + MAE 1시간 단위 예측

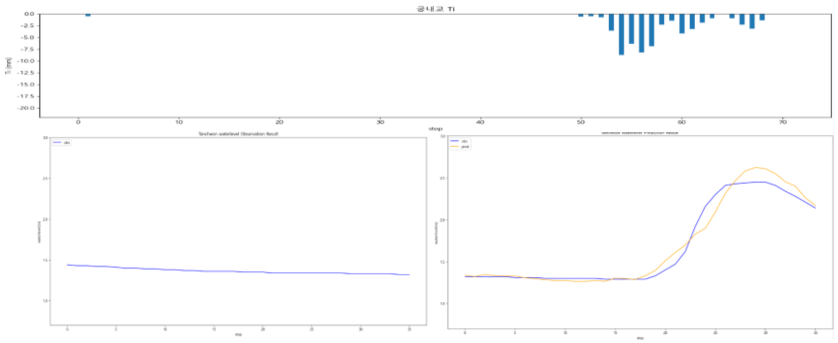
- 궁내교 : Lead time sensitivity 낮음 (시간 이동에 강건)
- 대곡교 : 궁내교보다 Lead-time sensitivity 큼 (특히 후행 1시간에서 성능 급락)
- peak 위치가 ±1h shift되어도 R^2 비슷하게 유지

		궁내교				대곡교			
Test data	time	r2	mse	mae	rmse	r2	mse	mae	rmse
24.07.23	17	0.6247	0.0329	0.0925	0.1816	0.6779	0.0171	0.1038	0.1311
	18	0.7308	0.0217	0.0962	0.1475	0.6749	0.0154	0.0860	0.1244
	19	0.7209	0.0217	0.0892	0.1475	0.4809	0.0191	0.1026	0.1385
25.07.16	16	0.7968	0.0283	0.1263	0.1684	0.9713	0.0168	0.1071	0.1298
	17	0.7947	0.0205	0.1047	0.1435	0.9470	0.0244	0.1098	0.1563
	18	0.8285	0.0174	0.1056	0.1319	0.9366	0.0201	0.1147	0.1419
25.08.14	04	0.7630	0.0277	0.1195	0.1665	0.8931	0.0489	0.1846	0.2211
	05	0.5087	0.0346	0.1446	0.1861	0.8781	0.0423	0.1589	0.2056
	06	0.6803	0.0131	0.0906	0.1146	0.6361	0.0707	0.2163	0.2659

궁내교 1시간 단위

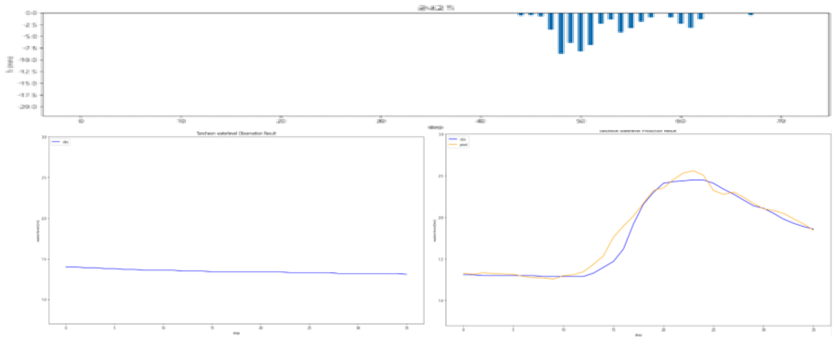
1시간 전 (01)

R2 : 0.9627



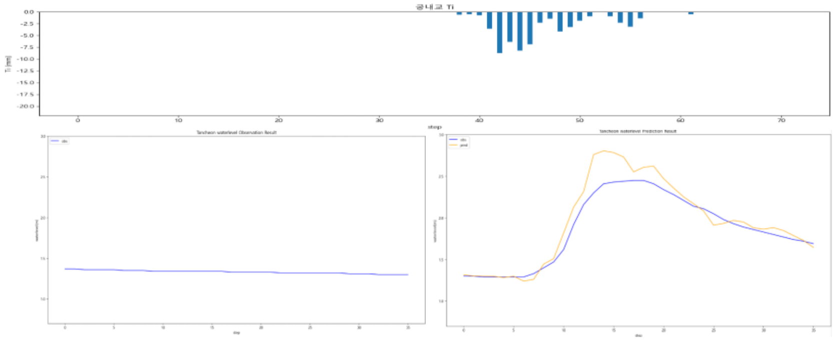
testdata_ 200802 (02)

R2 : 0.9656



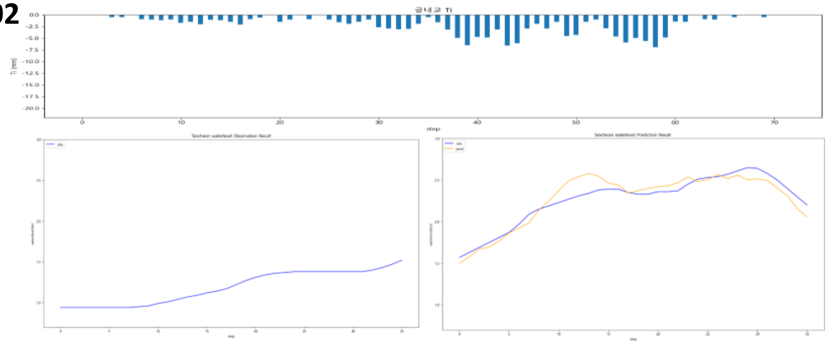
1시간 후 (03)

R2 : 0.8581



testdata_ 200802

R2 : 0.9656

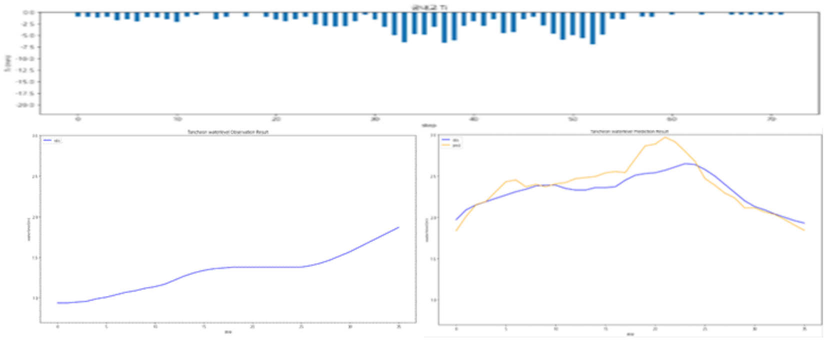


1시간 전 (15)

R2 : 0.8698

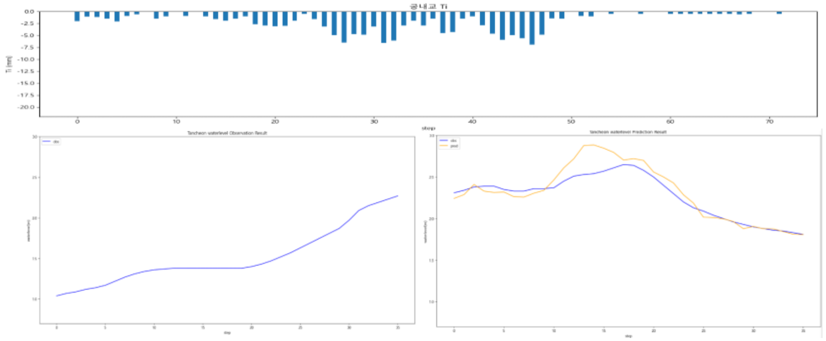
testdata_ 220712 (16)

R2 : 0.4201

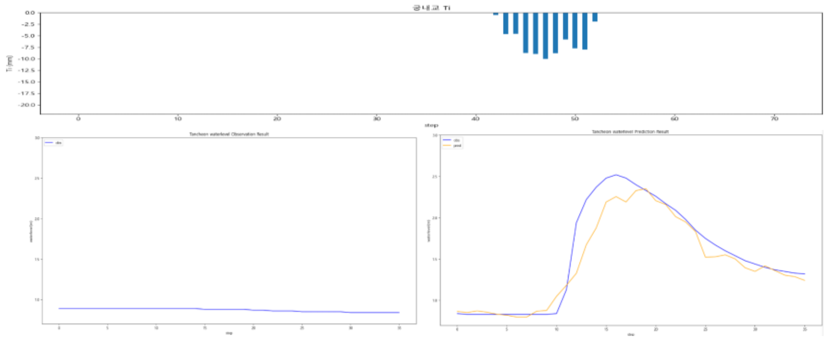


1시간 후 (17)

R2 : 0.7777

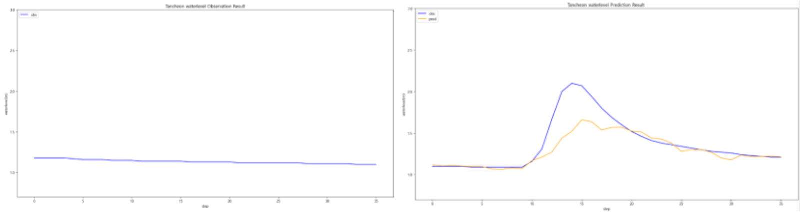


궁내교 1시간 단위



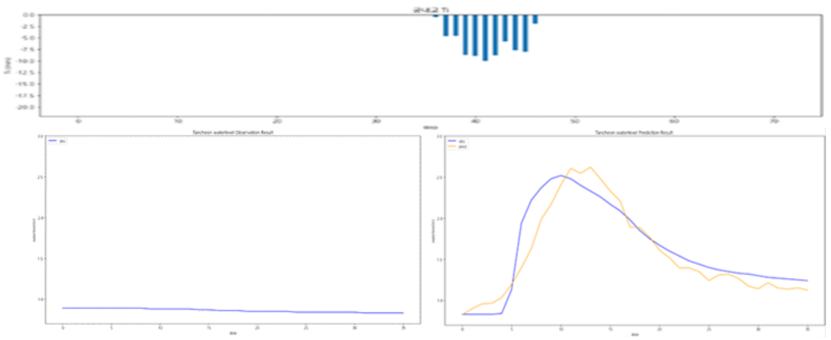
1시간 전 (06)

R2 : 0.8622



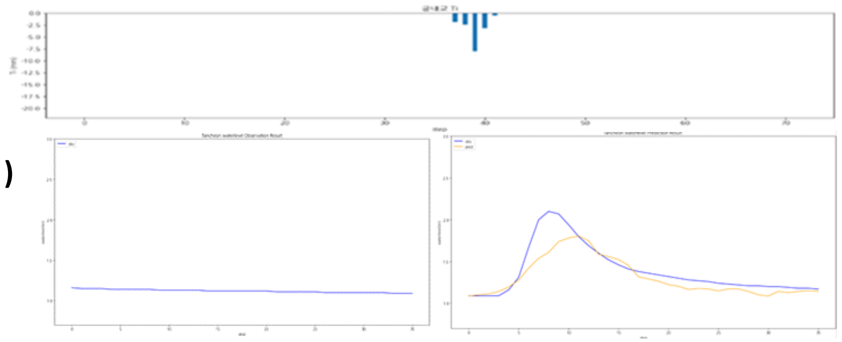
1시간 전 (17)

R2 : 0.6247



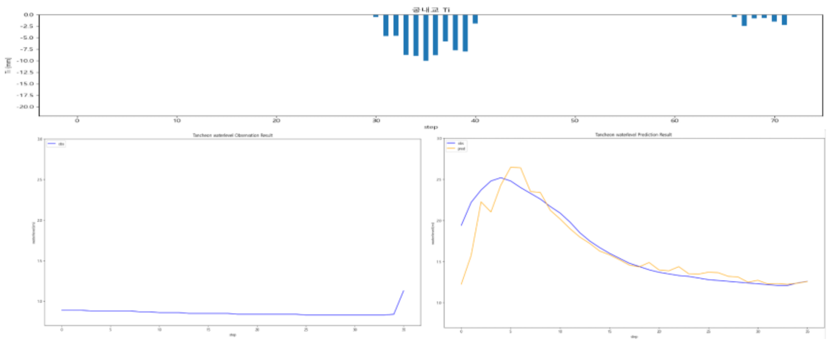
testdata_ 230711 (07)

R2 : 0.8622



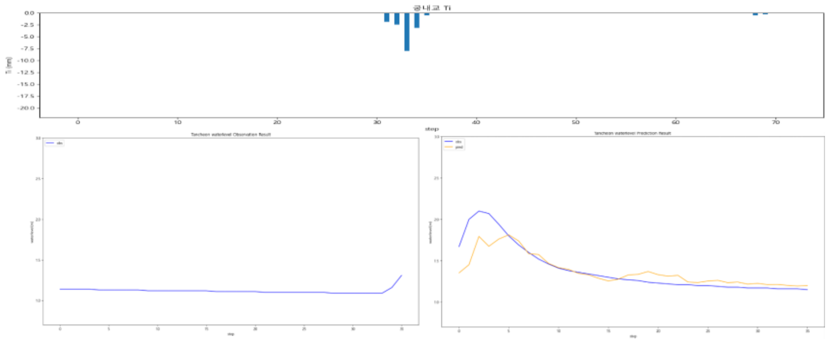
testdata_ 240723 (18)

R2 : 0.7308



1시간 후 (08)

R2 : 0.8323



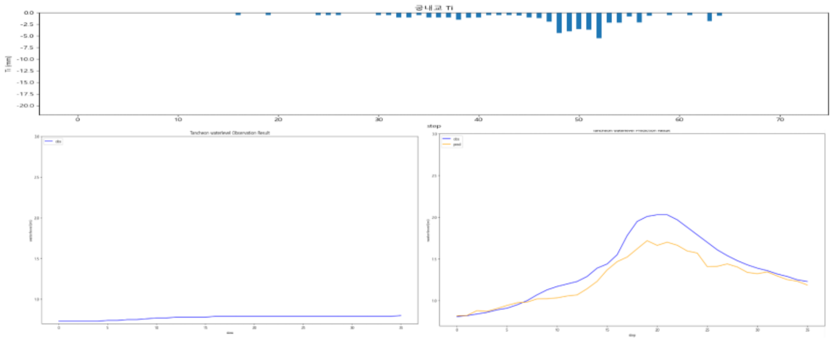
1시간 후 (19)

R2 : 0.7209

궁내교 1시간 단위

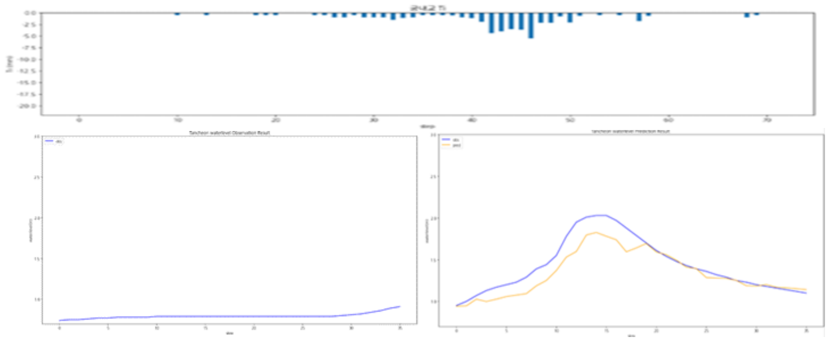
1시간 전 (16)

R2 : 0.7968



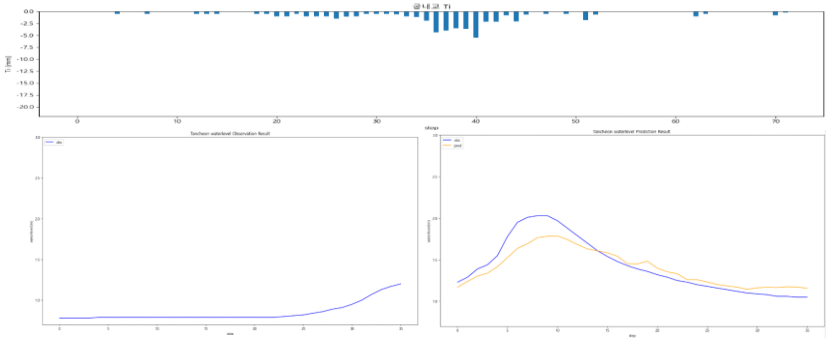
testdata_ 250716 (17)

R2 : 0.7947



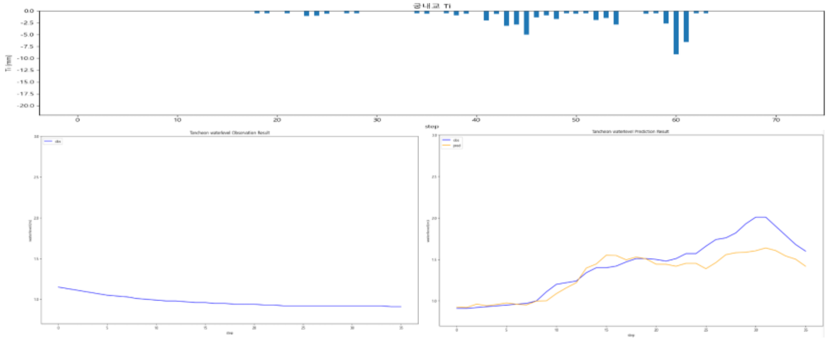
1시간 후 (18)

R2 : 0.8285



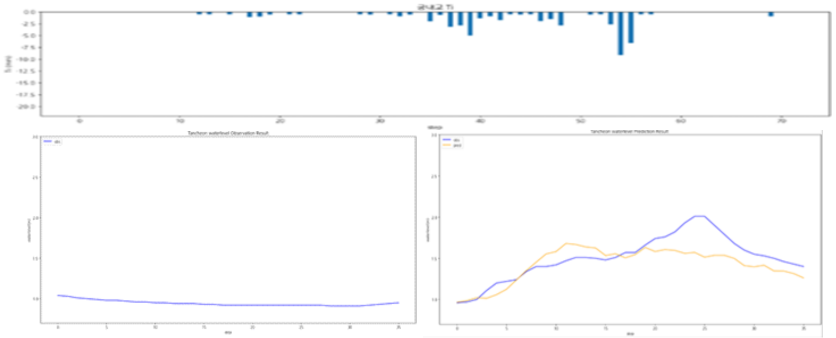
1시간 전 (04)

R2 : 0.7630



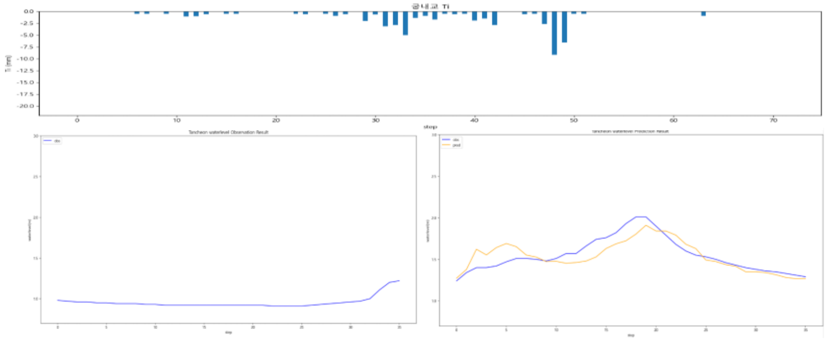
testdata_ 250814 (05)

R2 : 0.5087



1시간 후 (06)

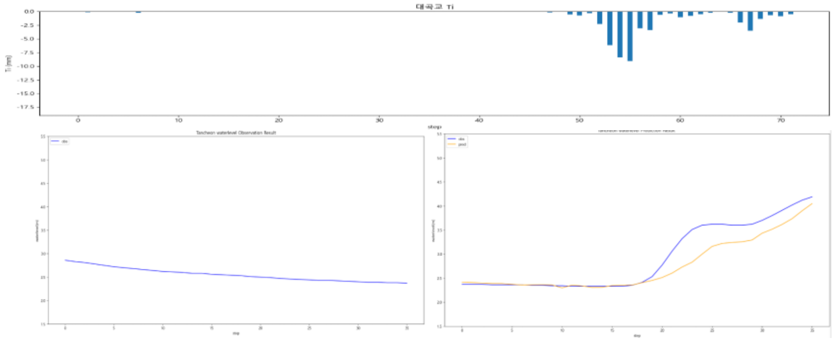
R2 : 0.6803



대곡교 - 1시간 단위

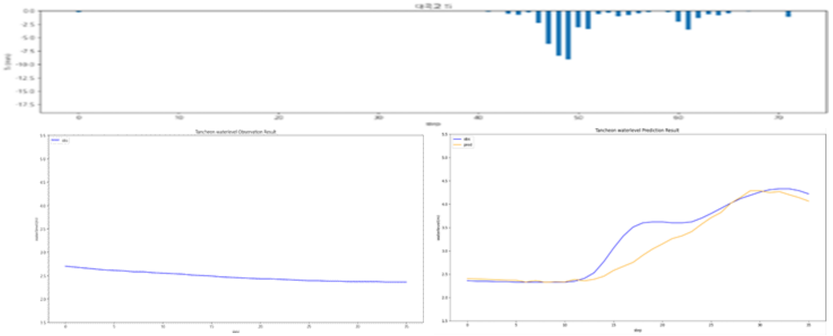
1시간 전 (01)

R2 : 0.8421



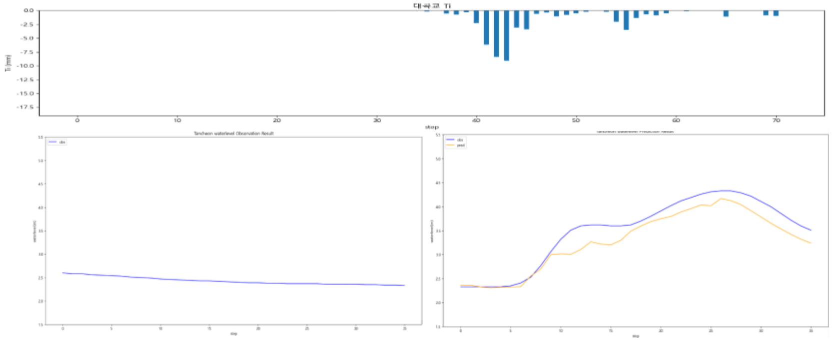
testdata_ 200802 (02)

R2 : 0.8748



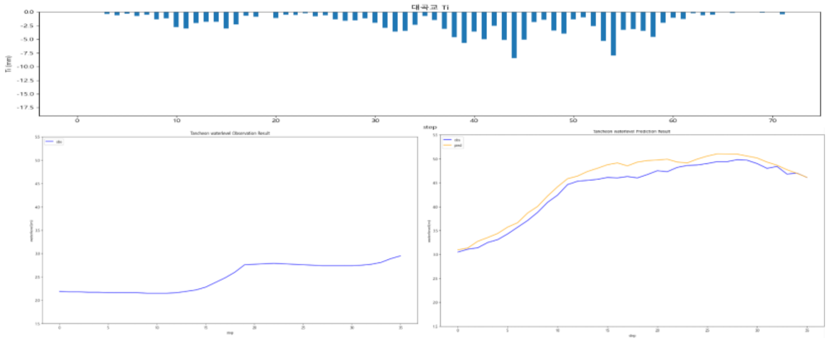
1시간 후 (03)

R2 : 0.8667



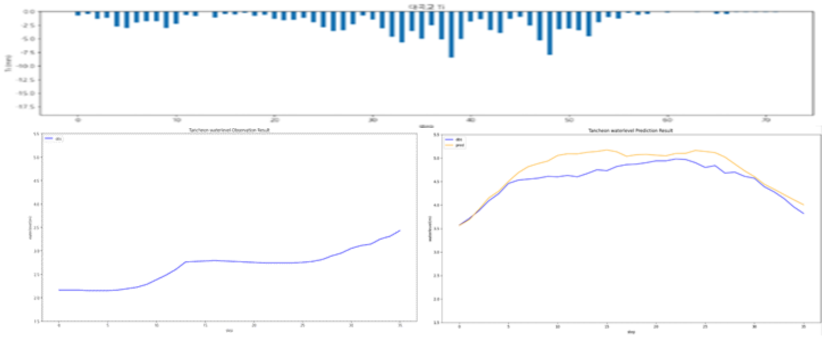
1시간 전 (15)

R2 : 0.9276



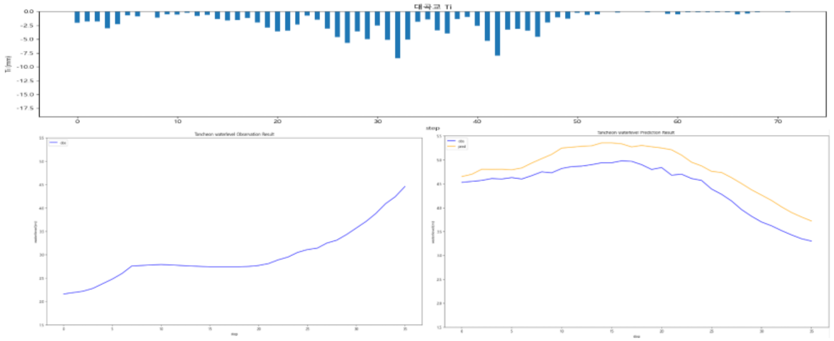
testdata_ 220712 (16)

R2 : 0.5503



1시간 후 (17)

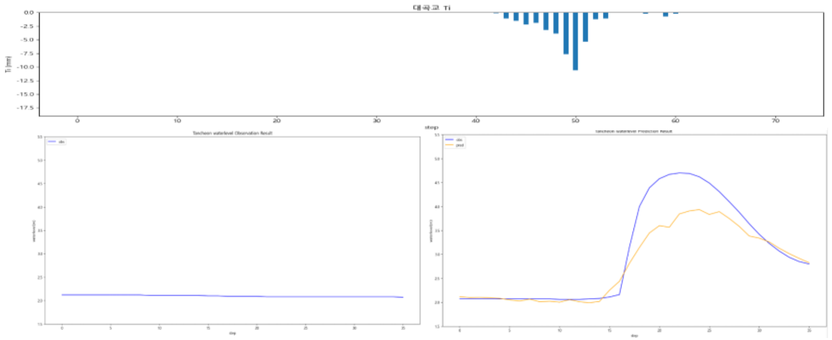
R2 : 0.3852



대곡교 - 1시간 단위

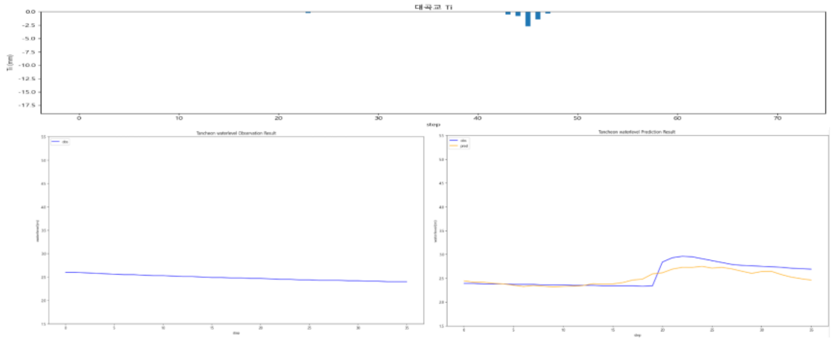
1시간 전 (06)

R2 : 0.8208



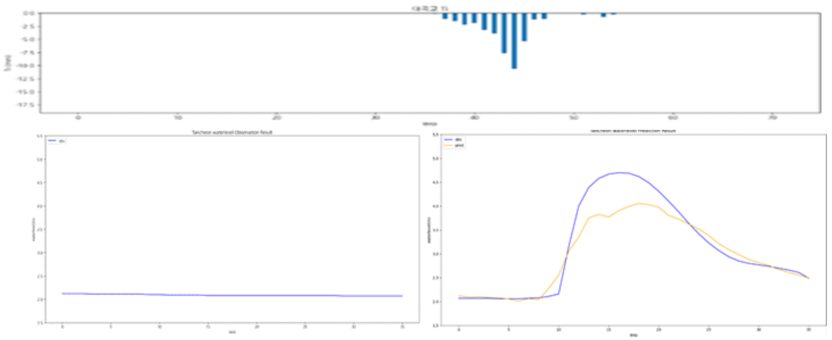
1시간 전 (17)

R2 : 0.6779



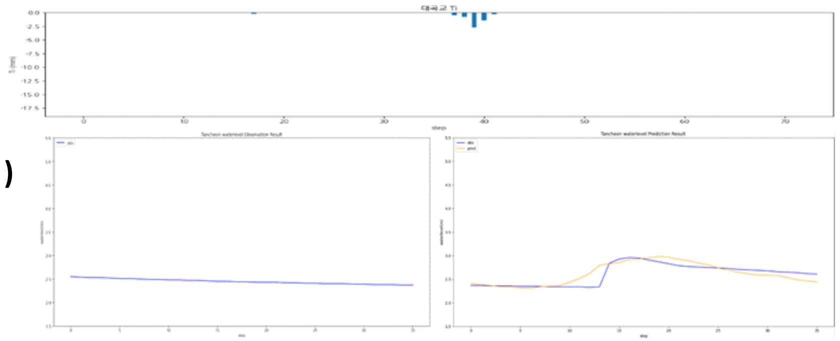
testdata_ 230711 (07)

R2 : 0.8659



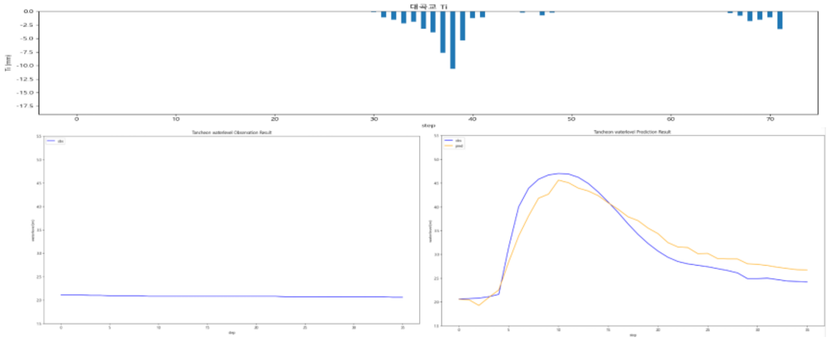
testdata_ 240723 (18)

R2 : 0.6749



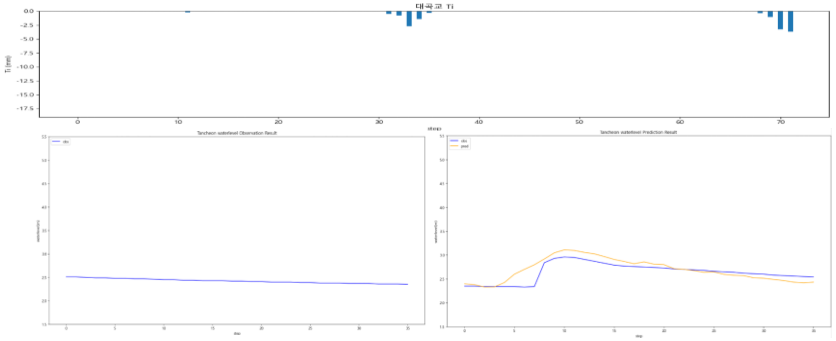
1시간 후 (08)

R2 : 0.9018



1시간 후 (19)

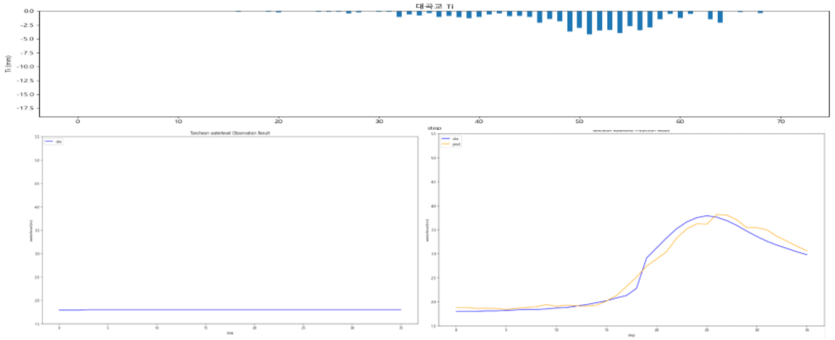
R2 : 0.4809



대곡교 - 1시간 단위

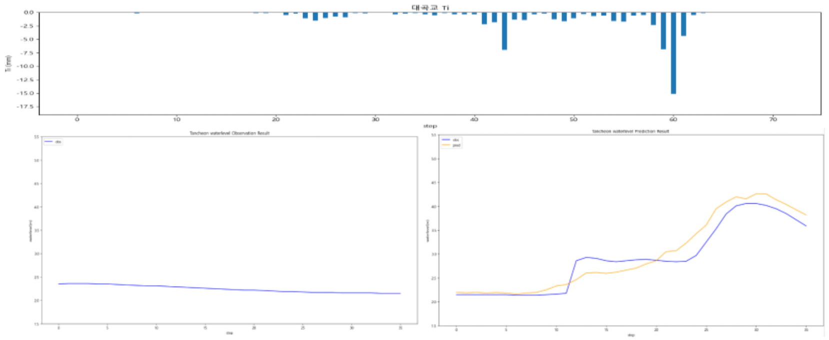
1시간 전 (16)

R2 : 0.9713



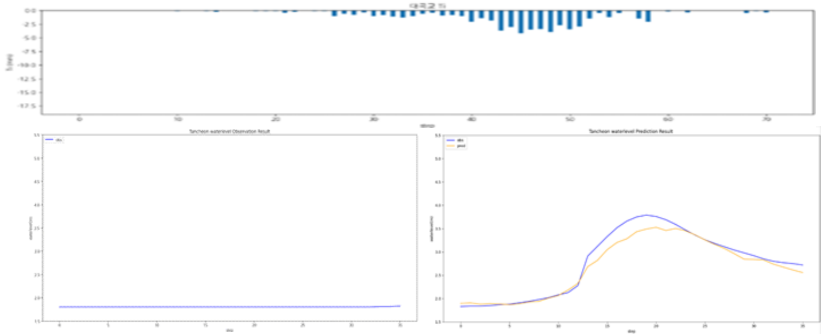
1시간 전 (04)

R2 : 0.8931



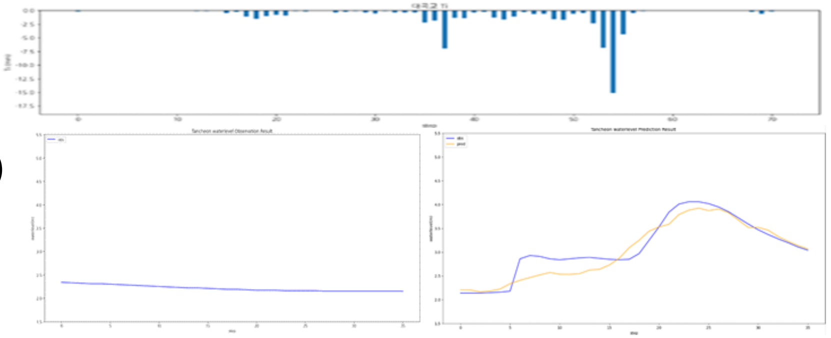
testdata_ 250716 (17)

R2 : 0.9470



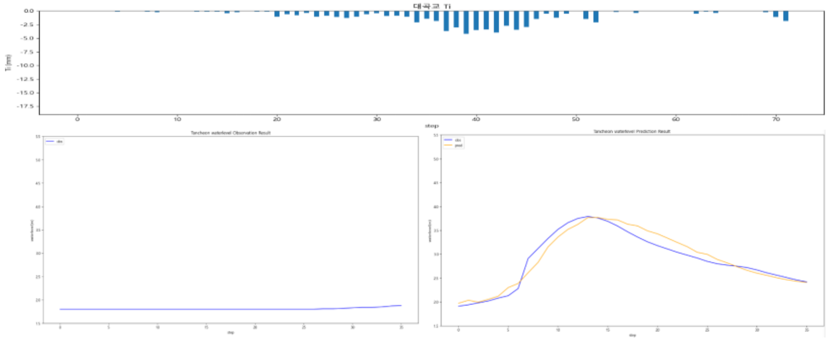
testdata_ 250814 (05)

R2 : 0.8781



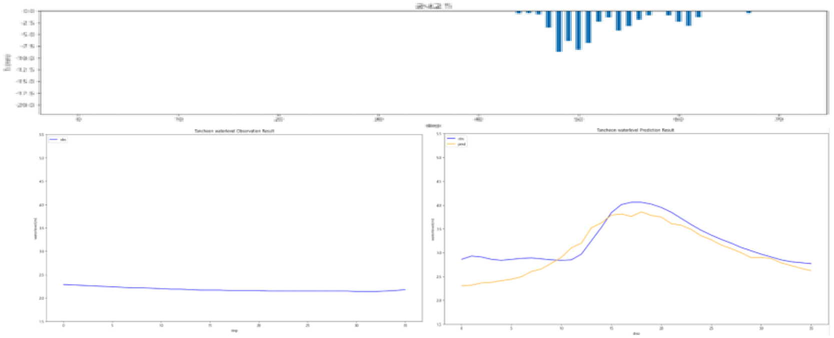
1시간 후 (18)

R2 : 0.9366



1시간 후 (06)

R2 : 0.6361



유량 예측 모델

학습데이터

- 관심 수위 : 궁내교 (2) 대곡교 (3.8)
- smoothing : 궁내교 (1) 대곡교 (2)
- 유량 그래프 스케일 고정 : 궁내교 (1.45 – 200.45) 대곡교 (3.93 – 712.55)
- 수위 그래프 스케일 고정 : 궁내교 (0.7 - 3) 대곡교 (1.5 – 5.5)

- learning_rate = 0.0005
- Loss_function = MAE
- Seq_length = 36 (6시간)
- Num_of_layer = 18

- 유량 곡선식 : 23년 11월 이후 ~

홍수통제소 API 호출 → MIET 활용 강우 이벤트 추출 → 관심 수위 이상 추출

time	서울시 (대곡교)	성남시 (성남북초교)	광주시 (남한산초교)	성남시 (대장동)	성남시 (구미초교)	성남시 (한국학중앙연구원)	성남시 (궁내교)_WL	성남시 (궁내교)_Q	서울시 (대곡교)_WL	서울시 (대곡교)_Q	대곡교_Ti	궁내교_Ti
2014-07-24 15:30:00	0.0	0.0	0.0	0.0	0.0	0.0	1.12	5.11	2.17	40.56	0.0	0.0

학습 , 검증 : 총 67개 中 61개 데이터 파일 사용

테스트 : 학습 데이터에서 제외한 6개 데이터 파일 (4개 이벤트)

+ 학습에서 검증에 사용된 2개 이벤트 사용

- 학습 : 2014 ~ 20230629 (43개)
- 검증 : 20230713 ~ 250813 (12개)
- 테스트 : 2550916 ~ 251006 (6개)

* model_(gn/dg)_flow

유량 예측 모델 (유량 비교)

- 평균적으로 안정적이나 특정 이벤트 성능 저하
- 궁내교 : 22.07.12 가 유독 안맞음 (R^2 음수)

기본 모델 대비 평균 R2 : 궁내교 ▼ 저하 (- 9%) , 대곡교 ▲ 개선 (+ 5.4%)

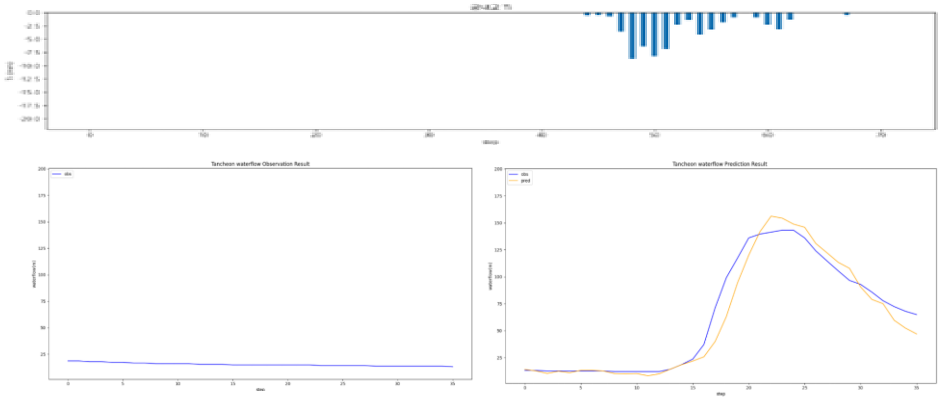
	궁내교				대곡교			
Test data	**r2**	**mse**	**mae**	**rmse**	**r2**	**mse**	**mae**	**rmse**
20.08.02	0.9470	133.2941	7.6087	11.5453	0.8104	2119.8319	31.4251	46.0416
22.07.12	-0.3192	1225.0217	27.96738	35.0003	0.9411	849.6534	22.2570	29.14881
23.07.11	0.9153	187.9712	10.4154	13.71025	0.7145	8592.75005	69.7216	92.6970
24.07.23	0.7854	99.3442	6.1220	9.9671	0.7932	92.4420	6.9991	9.6146
25.07.16	0.7869	113.8457	8.1333	10.6698	0.8734	916.8551	24.9128	30.2796
25.08.14	0.7481	88.8281	7.2954	9.4248	0.9158	666.7448	22.7927	25.8214

궁내교

* model_(gn/dg)_flow

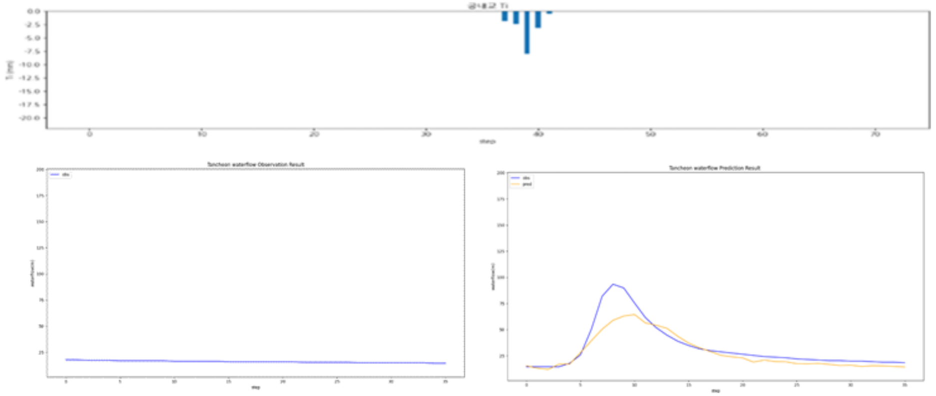
testdata_ 200802

R2 : 0.9470



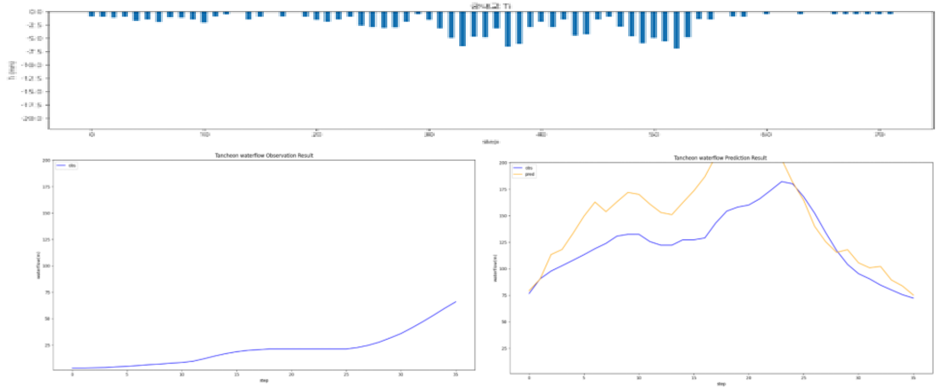
testdata_ 240723

R2 : 0.7854



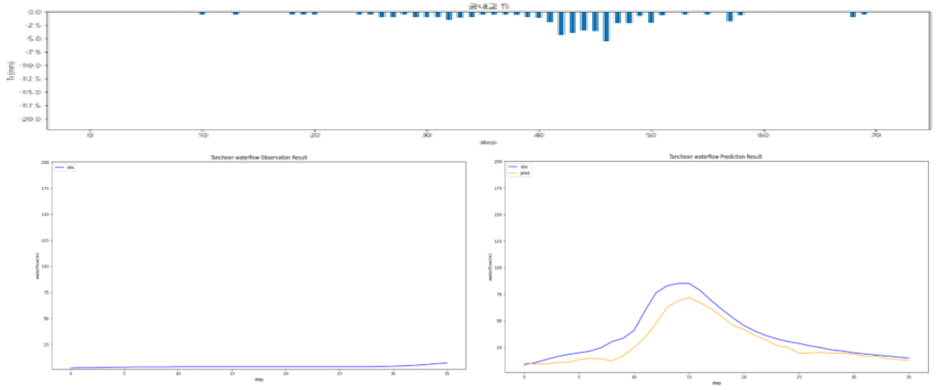
testdata_ 220712

R2 : -0.3192



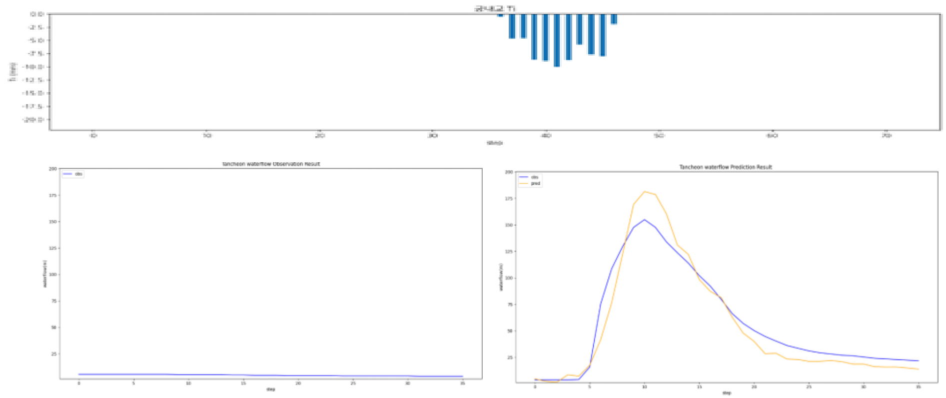
testdata_ 250716

R2 : 0.7869



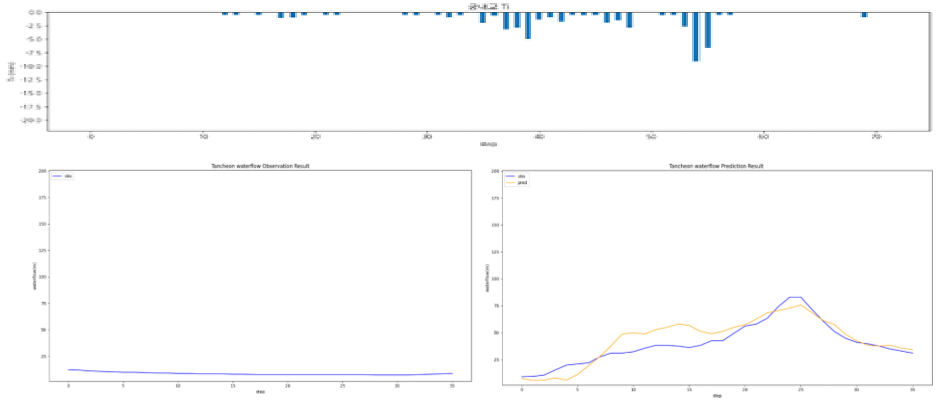
testdata_230711

R2 : 0.9153



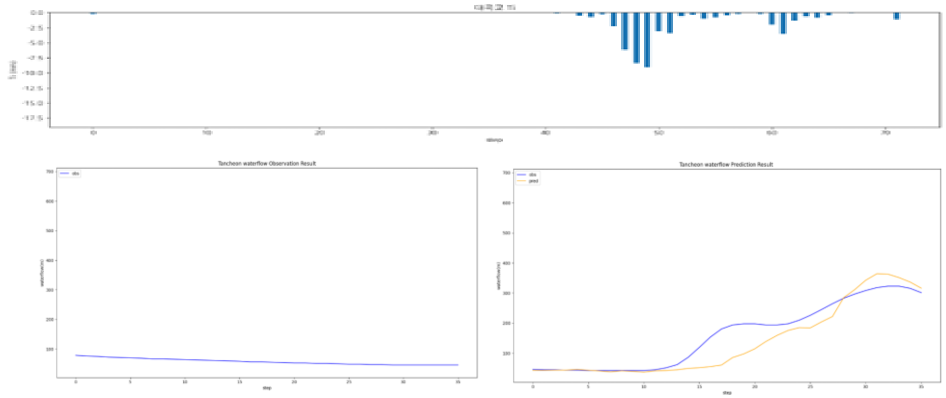
testdata_250814

R2 : 0.7481



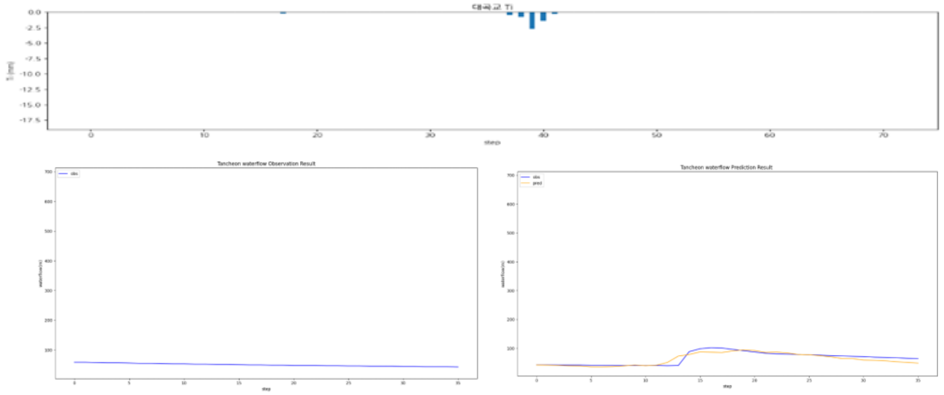
대곡교

* model_(gn/dg)_flow



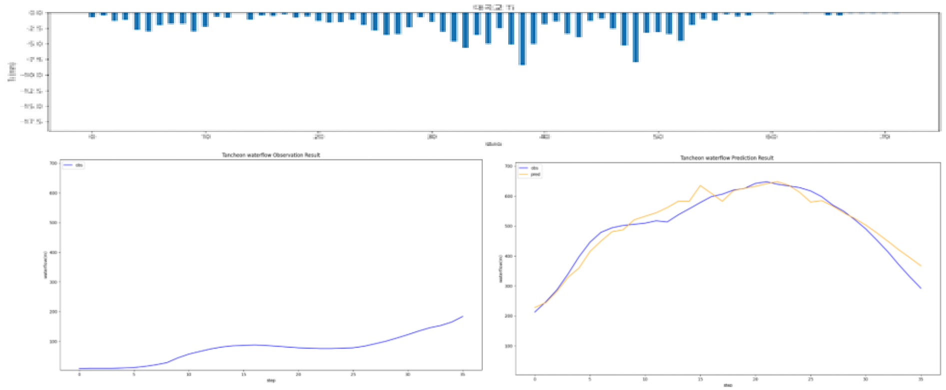
testdata_ 200802

R2 : 0.8104



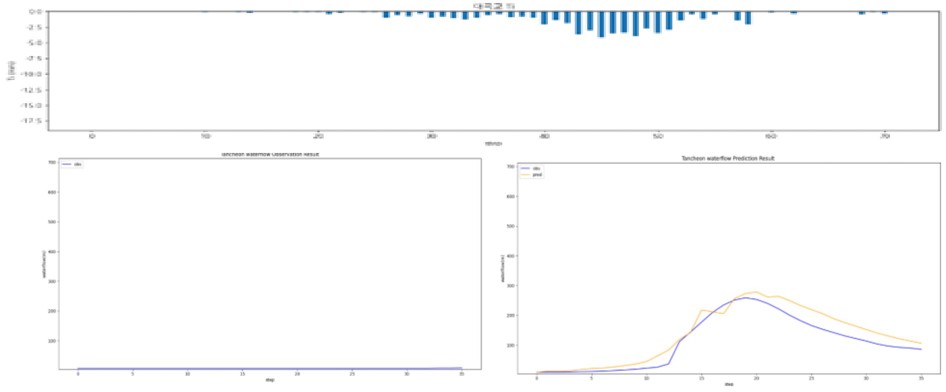
testdata_ 240723

R2 : 0.7932



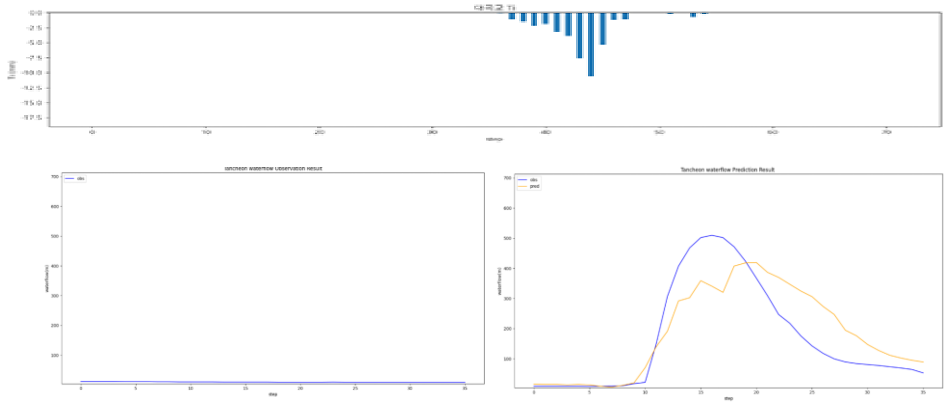
testdata_ 220712

R2 : 0.9411



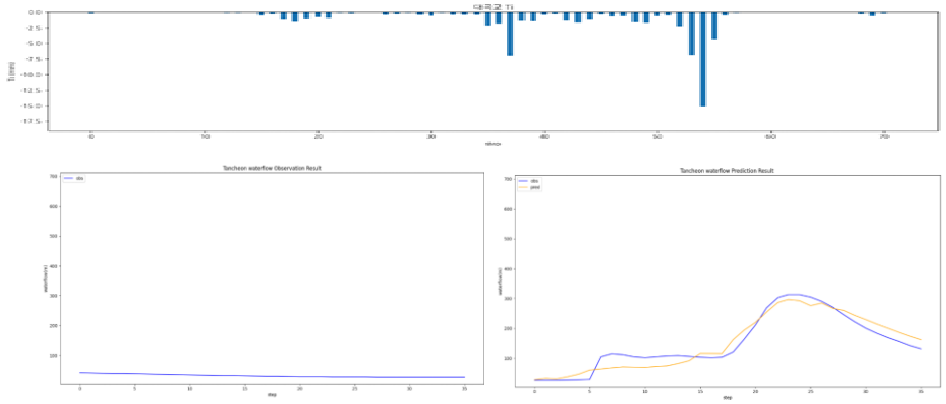
testdata_ 250716

R2 : 0.8734



testdata_230711

R2 : 0.7145



testdata_250814

R2 : 0.9158

대곡교 컬럼 제외

학습데이터

- 관심 수위 : 궁내교 (2) 대곡교 (3.8)
- smoothing : 궁내교 (1) 대곡교 (2)
- 그래프 스케일 고정 : 궁내교 (0.7 - 3) 대곡교 (1.5 – 5.5)

- learning_rate = 0.0005
- Loss_function = MAE
- Seq_length = 36 (6시간)
- Num_of_layer = 18

홍수통제소 API 호출 → MIET 활용 강우 이벤트 추출 → 관심 수위 이상 추출

time	성남시 (대장동)	성남시 (구미초교)	성남시 (한국학중앙연구원)	성남시 (궁내교)_WL	성남시 (궁내교)_Q	궁내교_Ti
2014-07-24 15:30:00	0.0	0.0	0.0	1.12	5.11	0.0

학습 , 검증 : 총 67개 中 60개 데이터 파일 사용

테스트 : 학습 데이터에서 제외한 7개 데이터 파일 (4개 이벤트)

+ 학습에서 검증에 사용된 2개 이벤트 사용

- 학습 : 2014 ~ 20230629 55번 (42개)
- 검증 : 20230629 56번 ~ 250813 (12개)
- 테스트 : 2550916 ~ 251006 (6개)

* model_(gn/dg)_none

대곡교 컬럼 제외

- 대곡교 정보 제거 → 궁내교 예측에도 악영향
- 상·하류 수위 상관성이 존재

기본 모델 대비 평균 R2 : 궁내교 ▼ 저하 (- 17.5%)

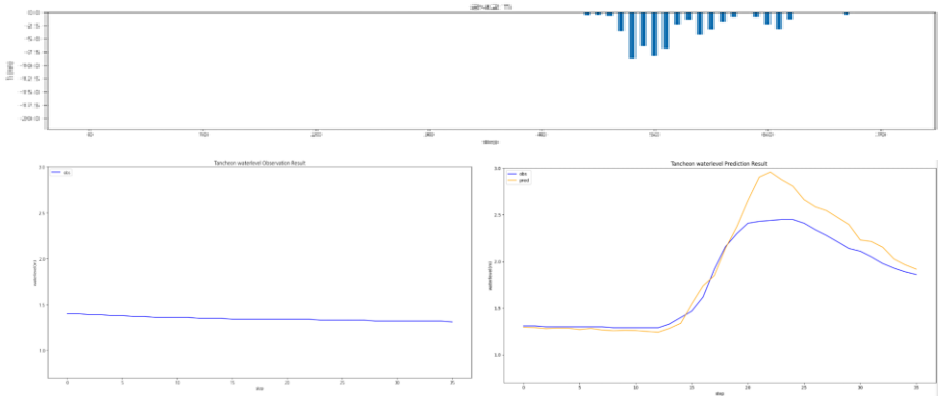
궁내교				
Test data	r2	mse	mae	rmse
20.08.02	0.8253	0.0369	0.1334	0.1922
22.07.12	-1.2836	0.0906	0.2571	0.3010
23.07.11	0.9267	0.0204	0.1150	0.1431
24.07.23	0.7720	0.0184	0.1050	0.1358
25.07.16	0.4997	0.0501	0.1578	0.2240
25.08.14	0.7909	0.0147	0.0918	0.1213

궁내교

*** model_(gn/dg)_wl18_3_2**

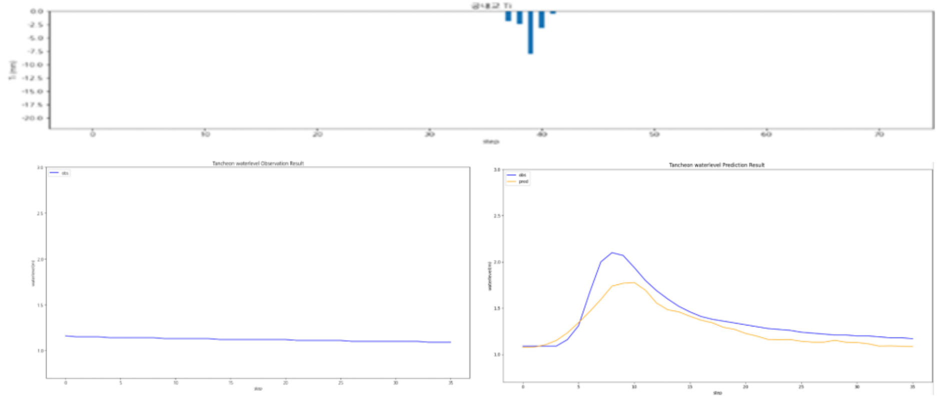
testdata_ 200802

R2 : 0.8253



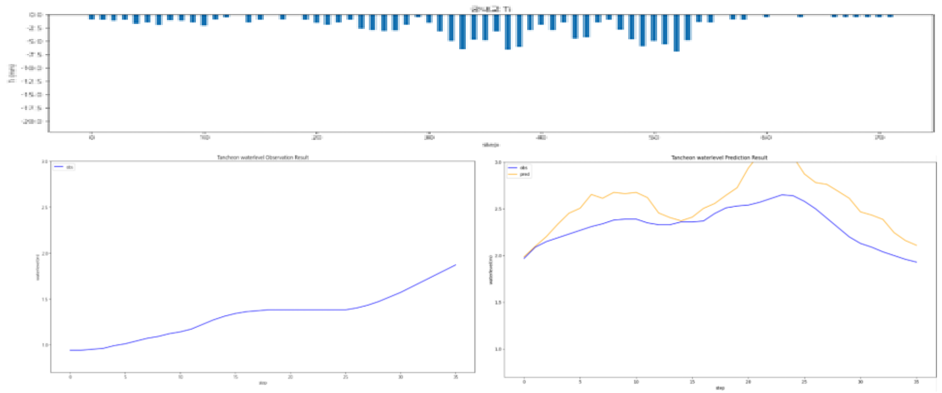
testdata_ 240723

R2 : 0.7720



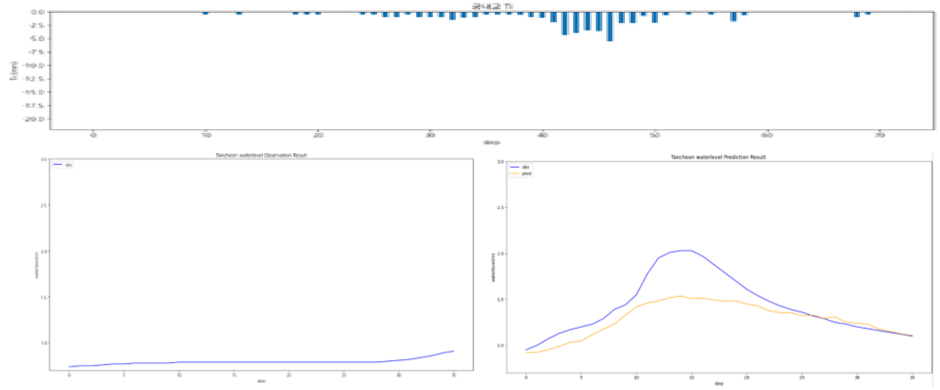
testdata_ 220712

R2 : -1.2836



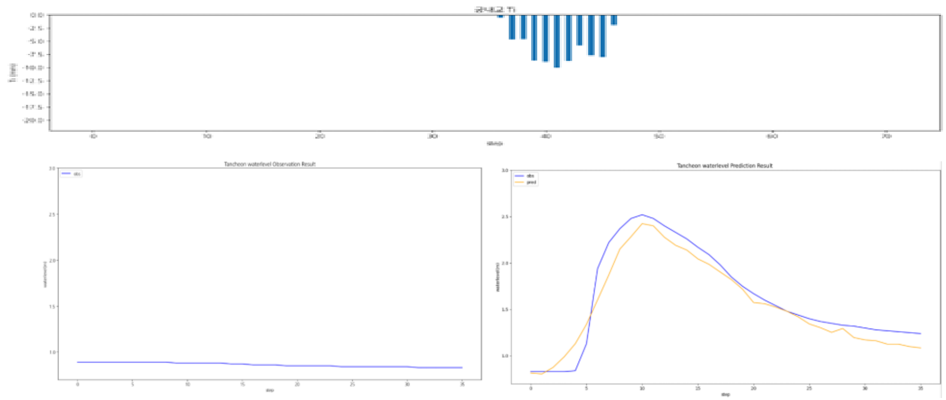
testdata_ 250716

R2 : 0.4997



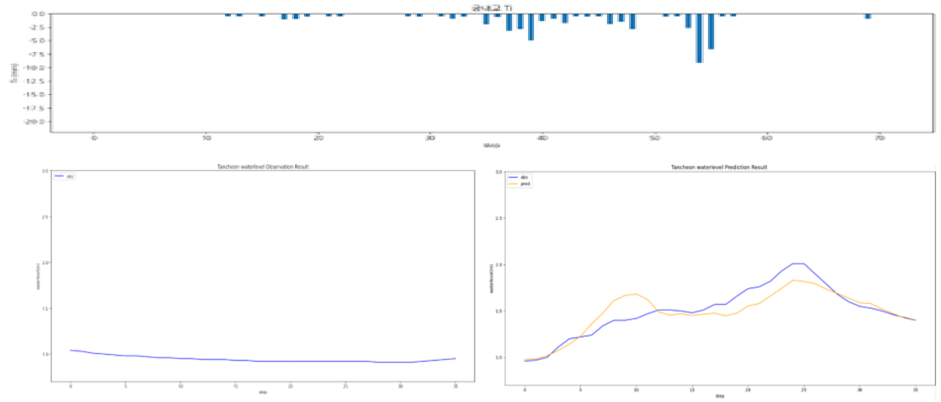
testdata_230711

R2 : 0.9267



testdata_250814

R2 : 0.7909



2020년 이후 데이터 학습

학습데이터

- 관심 수위 : 궁내교 (2) 대곡교 (3.8)
- smoothing : 궁내교 (1) 대곡교 (2)
- 그래프 스케일 고정 : 궁내교 (0.7 - 3) 대곡교 (1.5 – 5.5)

- learning_rate = 0.0005
- Loss_function = MAE
- Seq_length = 36 (6시간)
- Num_of_layer = 18

홍수통제소 API 호출 → MIET 활용 강우 이벤트 추출 → 관심 수위 이상 추출

time	서울시 (대곡교)	성남시 (성남북초교)	광주시 (남한산초교)	성남시 (대장동)	성남시 (구미초교)	성남시 (한국학중앙연구원)	성남시 (궁내교)_WL	성남시 (궁내교)_Q	서울시 (대곡교)_WL	서울시 (대곡교)_Q	대곡교_Ti	궁내교_Ti
2014-07-24 15:30:00	0.0	0.0	0.0	0.0	0.0	0.0	1.12	5.11	2.17	40.56	0.0	0.0

학습 , 검증 : 총 40개 中 34개 데이터 파일 사용

테스트 : 학습 데이터에서 제외한 6개 데이터 파일 (4개 이벤트)

+ 학습에서 검증에 사용된 2개 이벤트 사용

- 학습 : 2020 ~ 20240920 (24개)
- 검증 : 20240921 ~ 250919 85번 (7개)
- 테스트 : 250919 86번 ~ 251006 (3개)

* model_(gn/dg)_20

2020년 이후 데이터 학습

- 궁내교 : 최근 패턴 집중 → 소폭 개선
- 대곡교 : 학습 데이터 감소 → 일반화 성능 저하
- 지점별 데이터 분리 ???

기본 모델 대비 평균 R2 : 궁내교 ▲ 개선 (+ 2.5 %) , 대곡교 ▲ 개선 (+ 0.5 %)

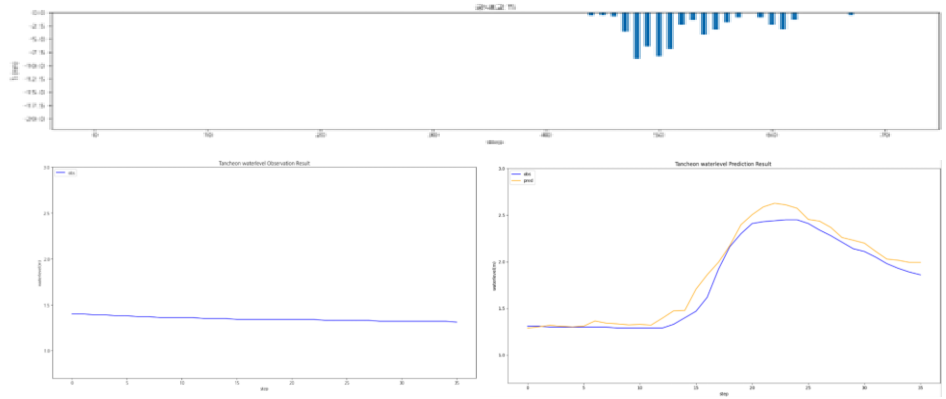
	궁내교				대곡교			
Test data	r2	mse	mae	rmse	r2	mse	mae	rmse
20.08.02	0.9513	0.0102	0.0814	0.1013	0.8845	0.0704	0.1797	0.2653
22.07.12	-0.8548	0.0735	0.2146	0.2712	-0.1561	0.1619	0.3550	0.4023
23.07.11	0.8751	0.0349	0.1217	0.1868	0.7304	0.2453	0.3701	0.4953
24.07.23	0.8136	0.0150	0.0610	0.1227	0.6077	0.0186	0.0981	0.1365
25.07.16	0.8213	0.0179	0.0927	0.1338	0.8647	0.0624	0.1862	0.2498
25.08.14	0.7835	0.0152	0.0984	0.1235	0.8840	0.0402	0.1572	0.2006

궁내교

* model_(gn/dg)_20

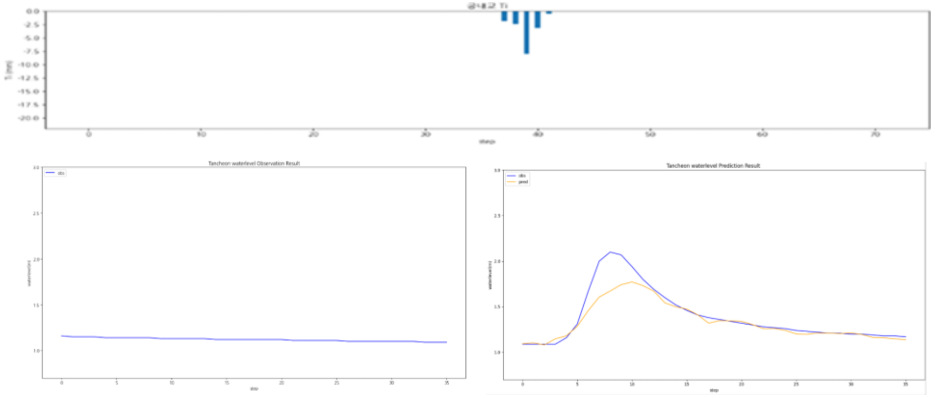
testdata_ 200802

R2 : 0.9513



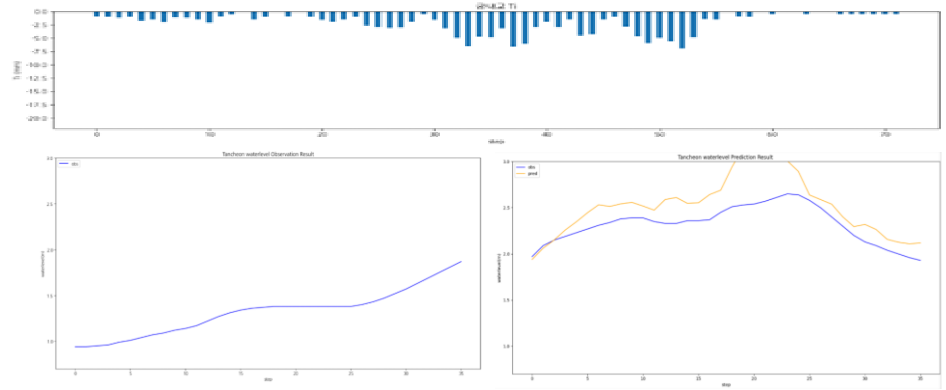
testdata_ 240723

R2 : 0.8136



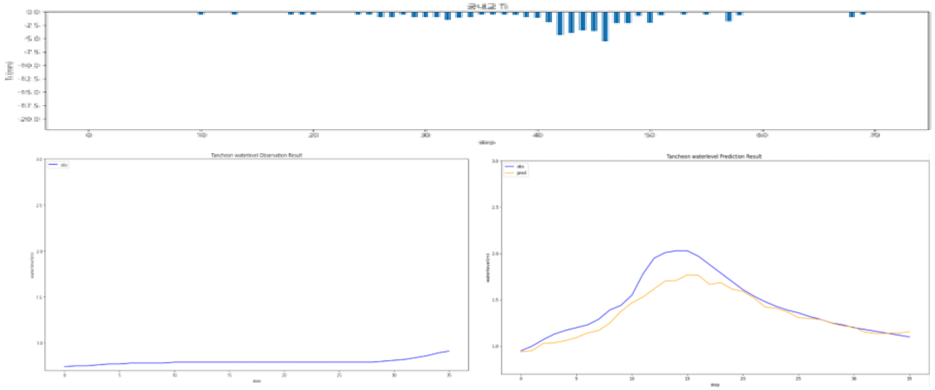
testdata_ 220712

R2 : -0.8548



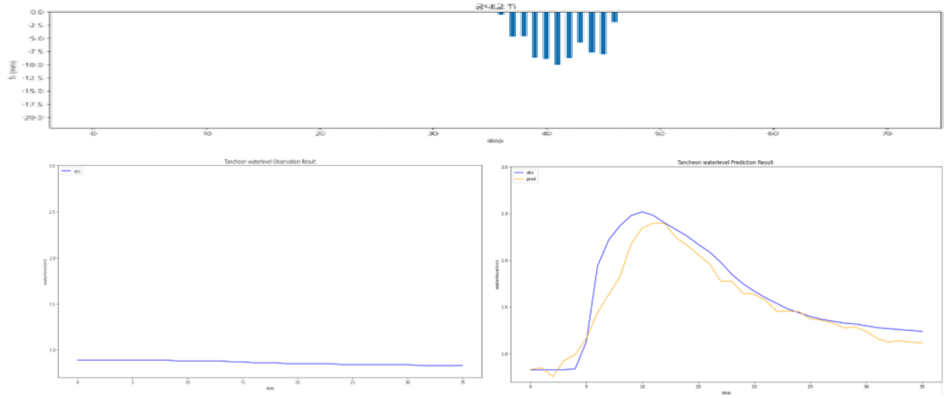
testdata_ 250716

R2 : 0.8213



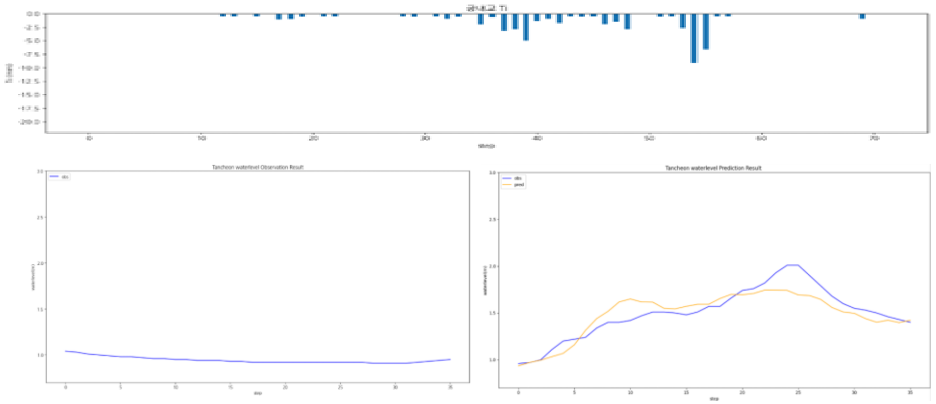
testdata_230711

R2 : 0.8751



testdata_250814

R2 : 0.7835

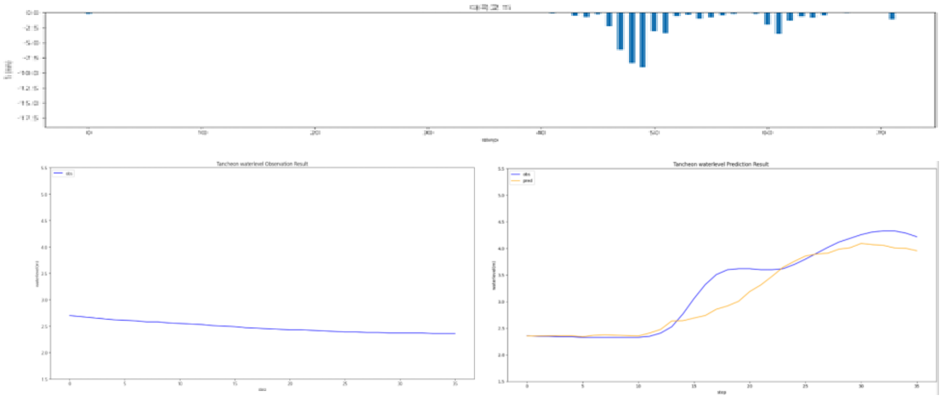


대곡교

* model_(gn/dg)_20

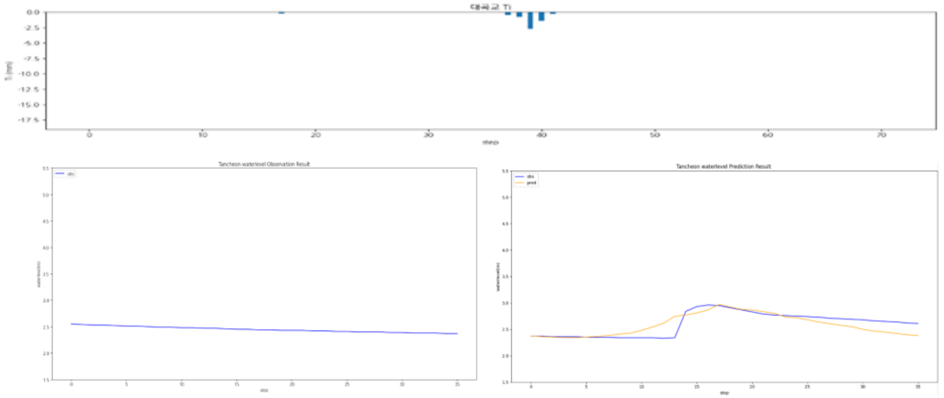
testdata_ 200802

R2 : 0.8845



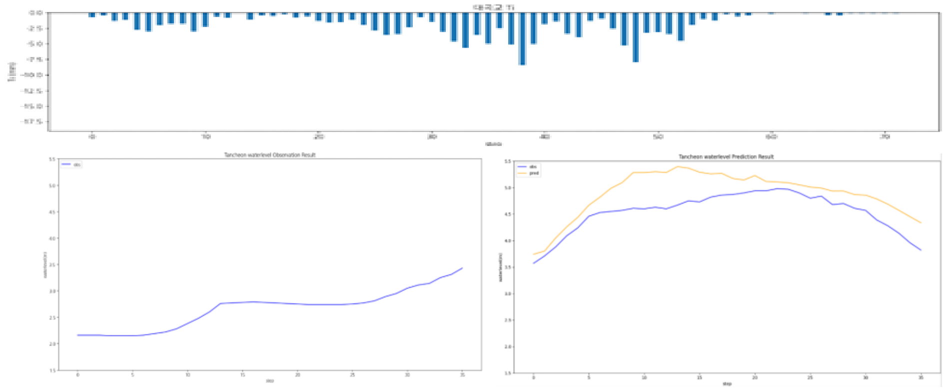
testdata_ 240723

R2 : 0.6077



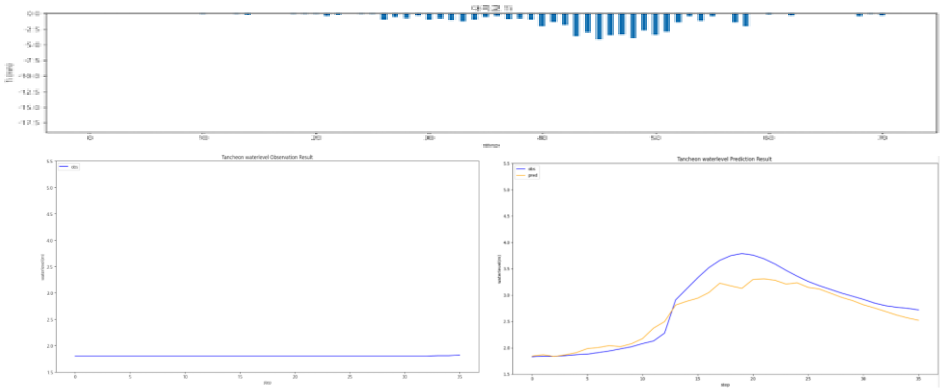
testdata_ 220712

R2 : -0.1561



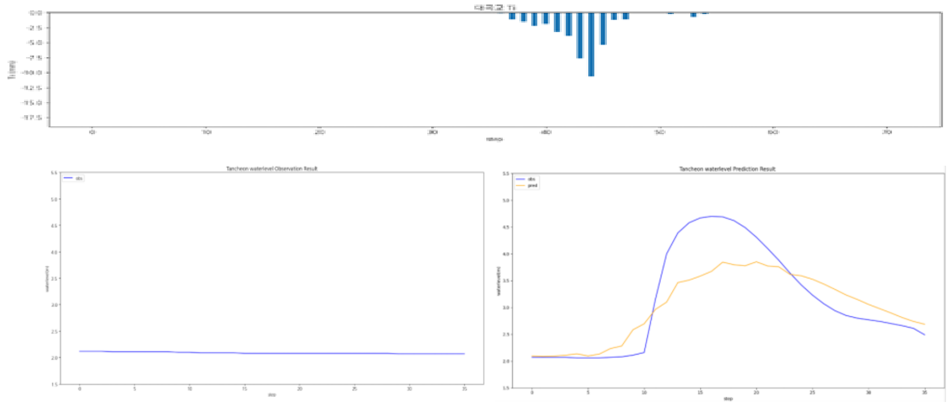
testdata_ 250716

R2 : 0.8647



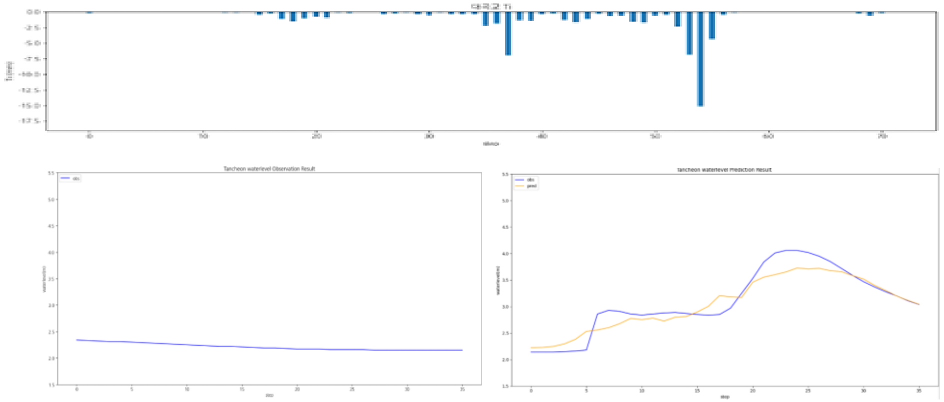
testdata_230711

R2 : 0.7304



testdata_250814

R2 : 0.8840



3시간 예측

학습데이터

- 관심 수위 : 궁내교 (2) 대곡교 (3.8)
- smoothing : 궁내교 (1) 대곡교 (2)
- 그래프 스케일 고정 : 궁내교 (0.7 - 3) 대곡교 (1.5 – 5.5)

- learning_rate = 0.0005
- Loss_function = MAE
- Seq_length = 36 (6시간)
- Num_of_layer = 18

홍수통제소 API 호출 → MIET 활용 강우 이벤트 추출 → 관심 수위 이상 추출

time	서울시 (대곡교)	성남시 (성남북초교)	광주시 (남한산초교)	성남시 (대장동)	성남시 (구미초교)	성남시 (한국학중앙연구원)	성남시 (궁내교)_WL	성남시 (궁내교)_Q	서울시 (대곡교)_WL	서울시 (대곡교)_Q	대곡교_Ti	궁내교_Ti
2014-07-24 15:30:00	0.0	0.0	0.0	0.0	0.0	0.0	1.12	5.11	2.17	40.56	0.0	0.0

학습 , 검증 : 총 67개 中 61개 데이터 파일 사용

테스트 : 학습 데이터에서 제외한 6개 데이터 파일 (4개 이벤트)

+ 학습에서 검증에 사용된 2개 이벤트 사용

- 학습 : 2014 ~ 20230629 (43개)
- 검증 : 20230713 ~ 250813 (12개)
- 테스트 : 2550916 ~ 251006 (6개)

* model_(gn/dg)_3

3시간 예측

- 이벤트별 변동성 가장 큼
- 대곡교에서 -4.21 R^2 발생
- R^2 기준으로는 손해

기본 모델 대비 평균 R2 : 궁내교 ▼ 저하 (- 13.4%) , 대곡교 ▼ 저하 (-26.5%)

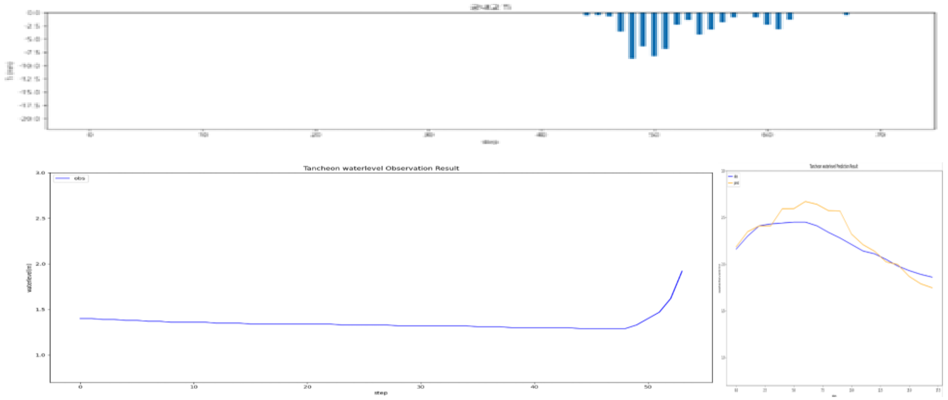
	궁내교				대곡교			
Test data	**r2**	**mse**	**mae**	**rmse**	**r2**	**mse**	**mae**	**rmse**
20.08.02	0.5417	0.0184	0.1049	0.1359	0.4341	0.0495	0.1670	0.2225
22.07.12	0.8669	0.0084	0.0795	0.0921	0.9659	0.0042	0.0452	0.0651
23.07.11	0.9006	0.0031	0.0452	0.0563	0.7131	0.1341	0.3316	0.3663
24.07.23	0.5351	0.0014	0.0333	0.0384	-4.2174	0.0347	0.1570	0.1863
25.07.16	0.6605	0.0135	0.0984	0.1165	0.7288	0.0379	0.1619	0.1947
25.08.14	0.2037	0.0287	0.1493	0.1695	0.8953	0.0140	0.0854	0.1186

궁내교

* model_(gn/dg)_3

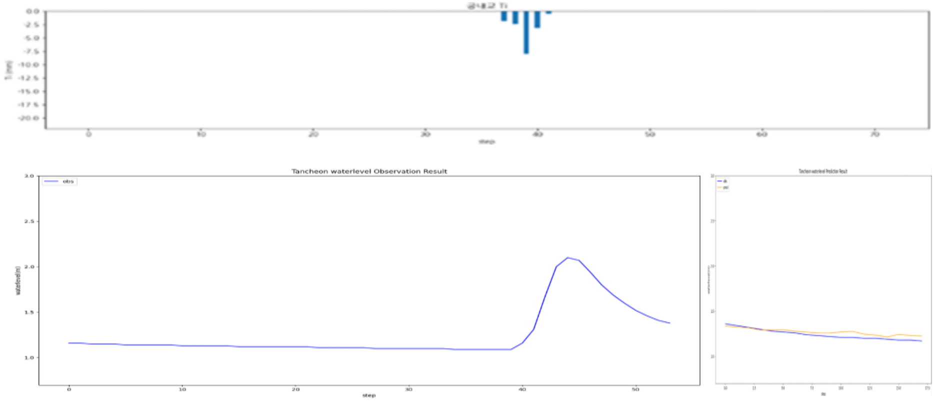
testdata_ 200802

R2 : 0.5417



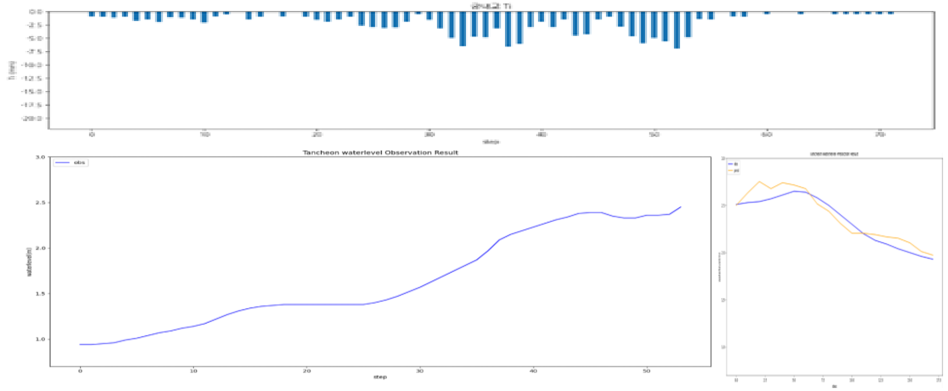
testdata_ 240723

R2 : 0.5351



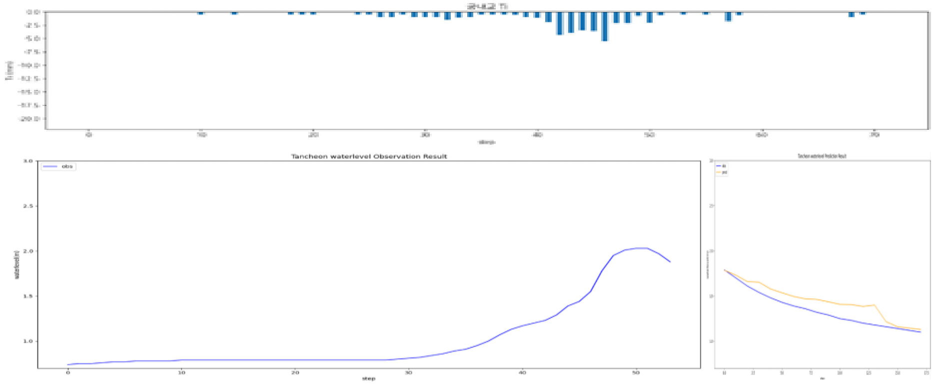
testdata_ 220712

R2 : 0.8669



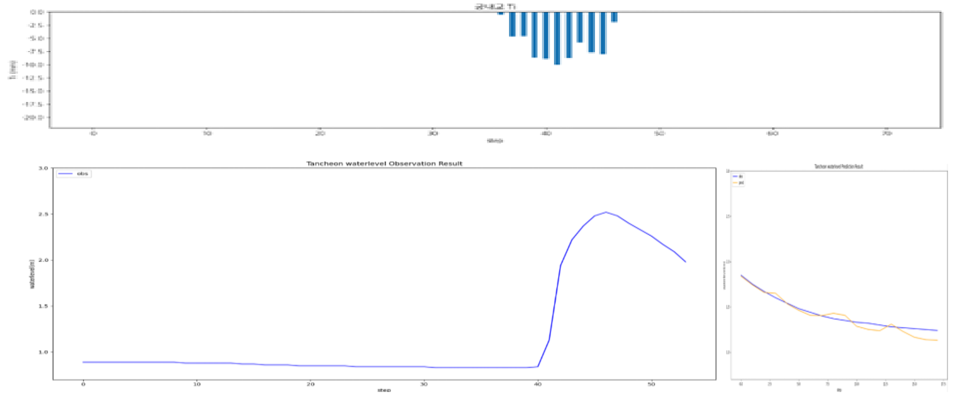
testdata_ 250716

R2 : 0.6605



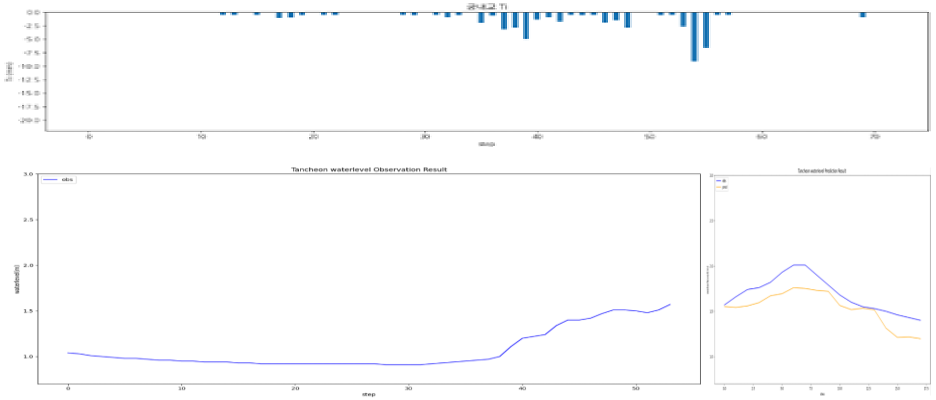
testdata_230711

R2 : 0.9006



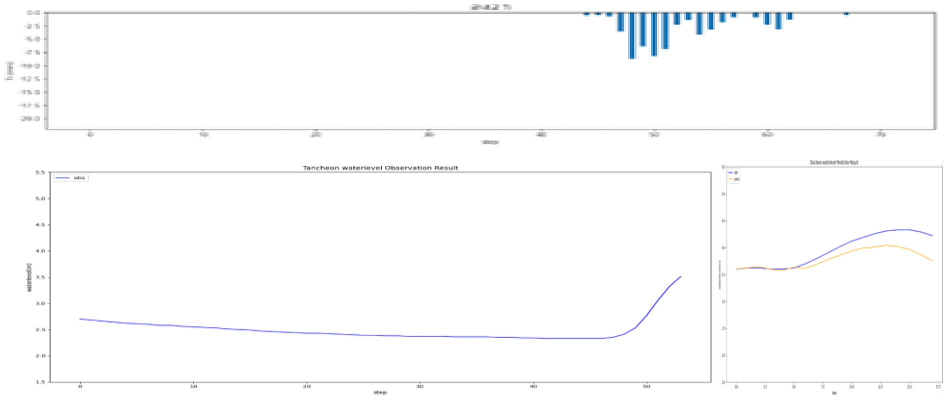
testdata_250814

R2 : 0.2037



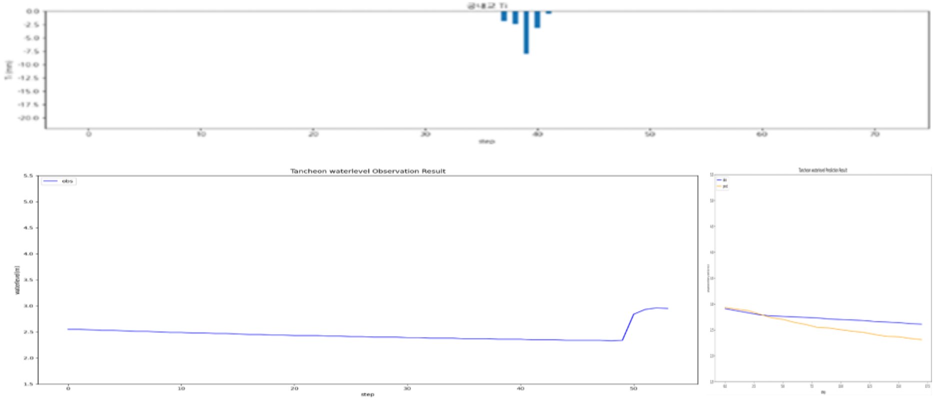
대곡교

* model_(gn/dg)_3



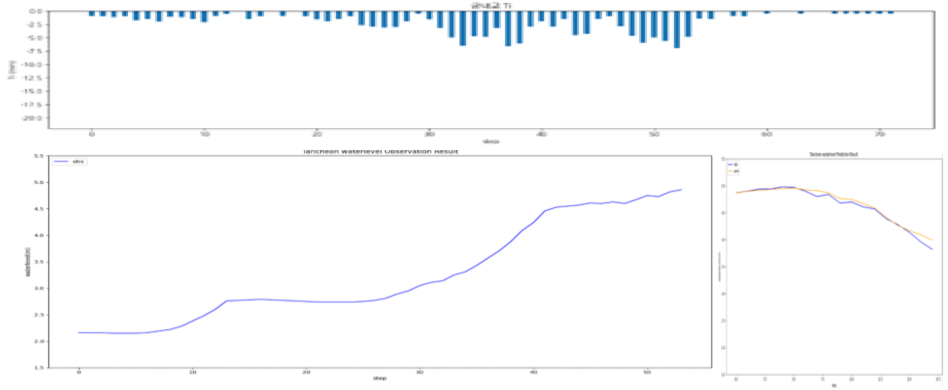
testdata_ 200802

R2 : 0.4341



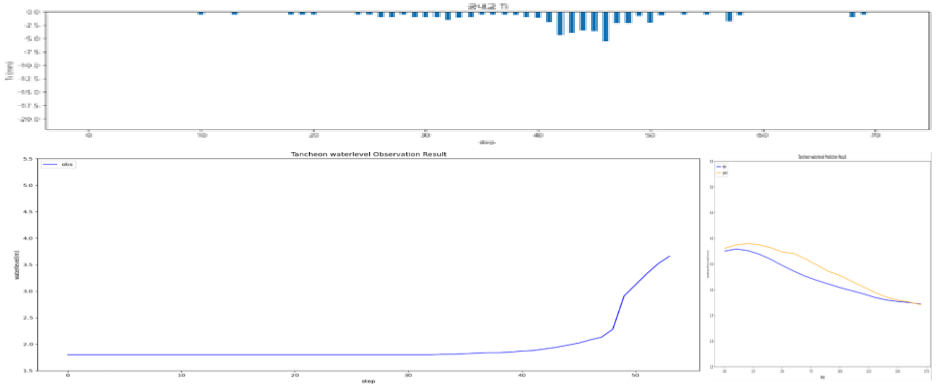
testdata_ 240723

R2 : -4.2174



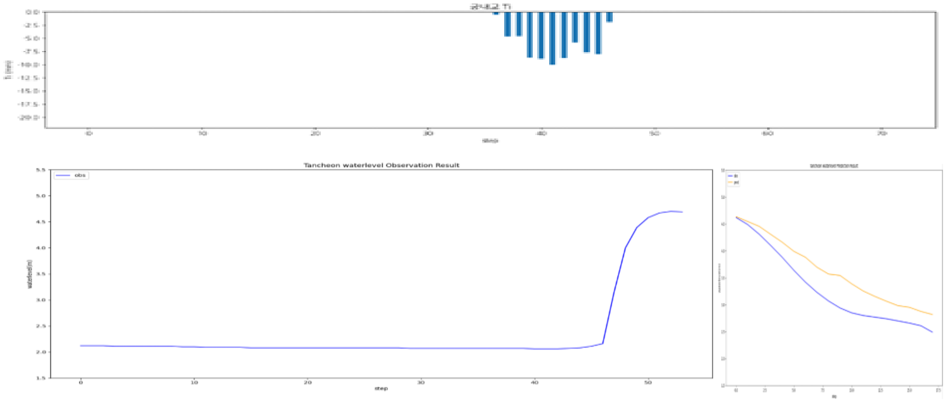
testdata_ 220712

R2 : 0.9659



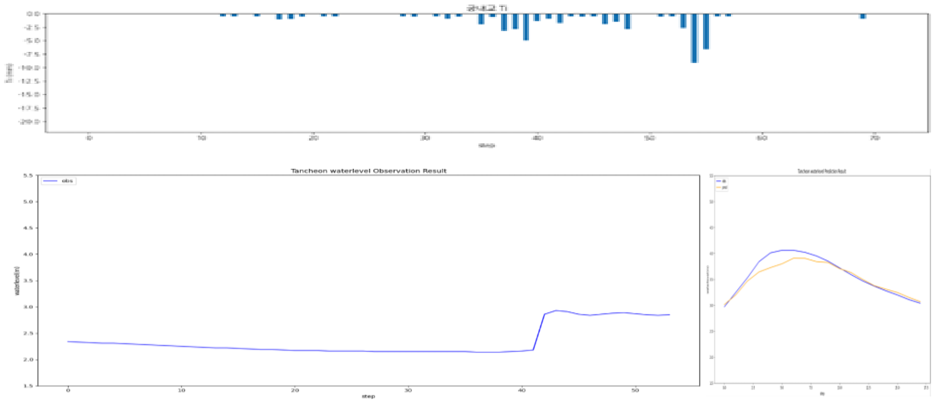
testdata_ 250716

R2 : 0.7288



testdata_230711

R2 : 0.7131



testdata_250814

R2 : 0.8953

* model_(gn/dg)_3

3시간 예측 (예측 시작 시점 -1시간)

- 이벤트별 변동성 가장 큼
- 대곡교에서 -4.21 R^2 발생
- R^2 기준으로는 손해

기본 모델 대비 평균 R2 : 궁내교 ▼ 저하 (- 13.4%) , 대곡교 ▼ 저하 (-26.5%)

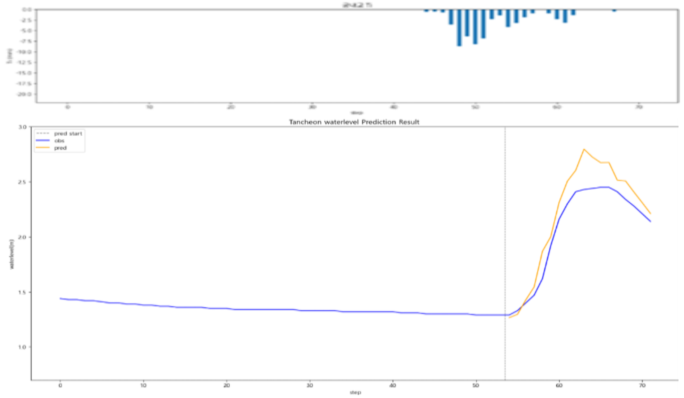
	궁내교				대곡교			
Test data	r2	mse	mae	rmse	r2	mse	mae	rmse
20.08.02	0.8214	0.0315	0.1508	0.1774	0.9113	0.0216	0.1275	0.1470
22.07.12	-0.3291	0.0215	0.1190	0.1467	-0.1308	0.0159	0.1159	0.1264
23.07.11	0.9196	0.0107	0.0819	0.1035	0.7288	0.1252	0.3182	0.3538
24.07.23	0.7568	0.0126	0.0911	0.1122	0.5679	0.0126	0.0911	0.1122
25.07.16	0.9456	0.0045	0.0528	0.0671	0.5949	0.0571	0.2051	0.2390
25.08.14	-0.9700	0.0607	0.2157	0.2464	0.8942	0.0257	0.1387	0.1604

궁내교

* model_(gn/dg)_3

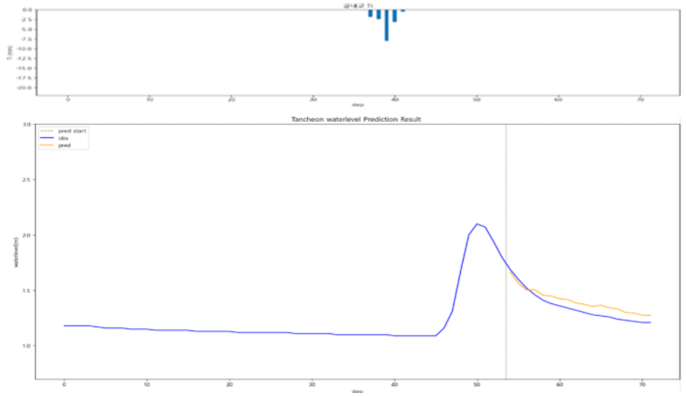
testdata_ 200802

R2 : 0.8214



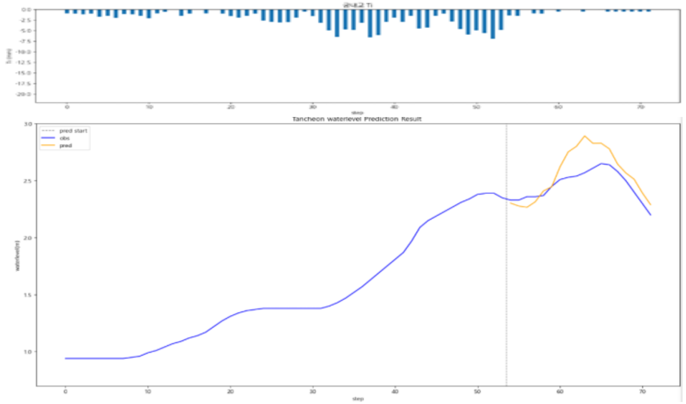
testdata_ 240723

R2 : 0.7568



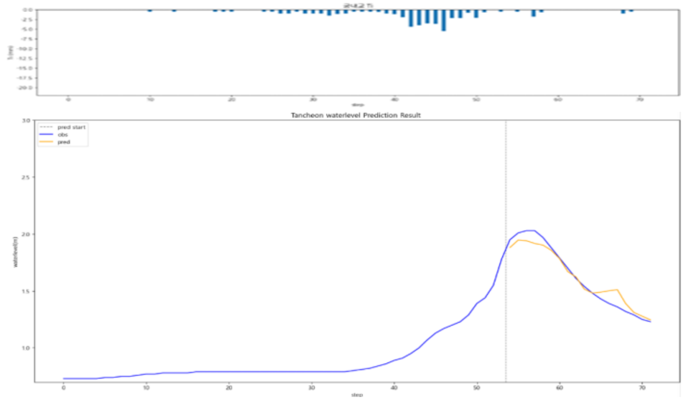
testdata_ 220712

R2 : -0.3291



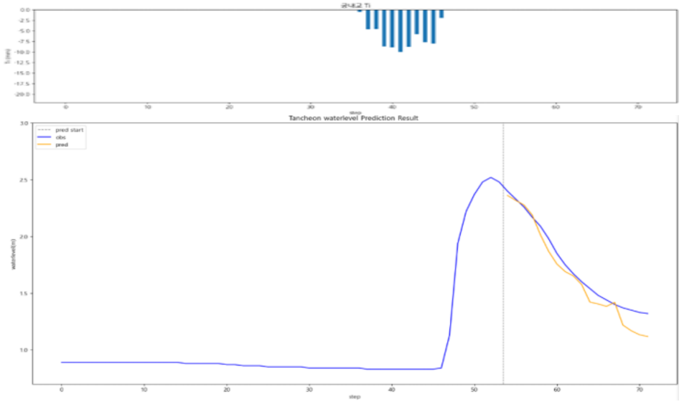
testdata_ 250716

R2 : 0.9456



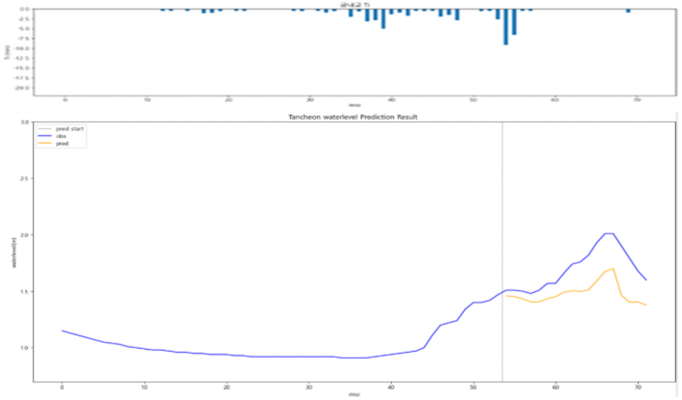
testdata_230711

R2 : 0.9196



testdata_250814

R2 : -0.9700

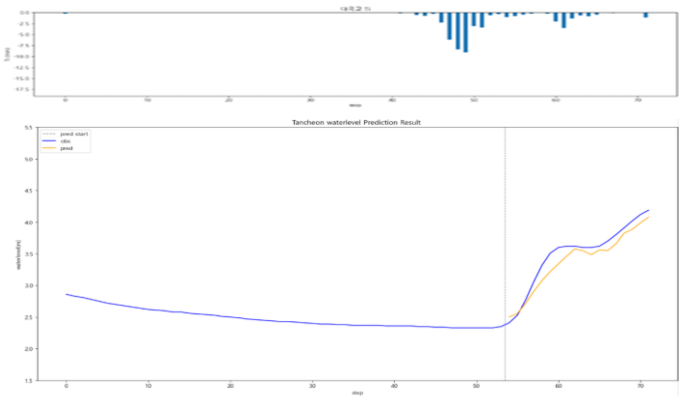


대곡교

* model_(gn/dg)_3

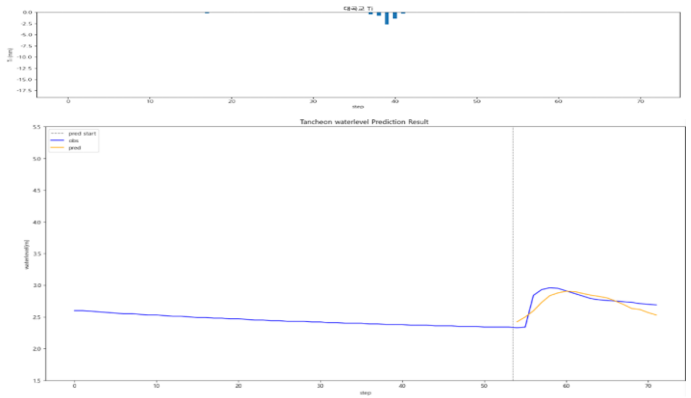
testdata_ 200802

R2 : 0.9113



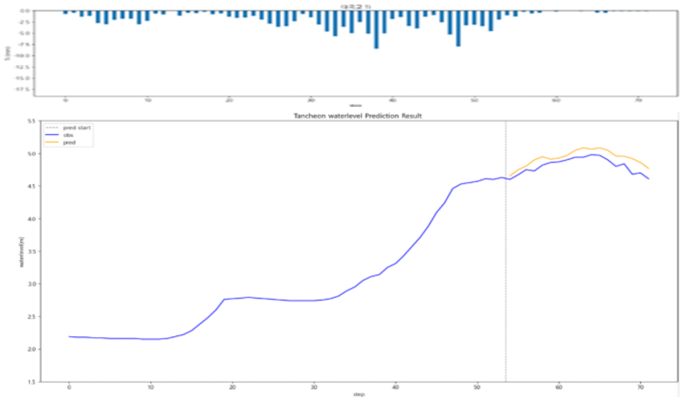
testdata_ 240723

R2 : 0.5679



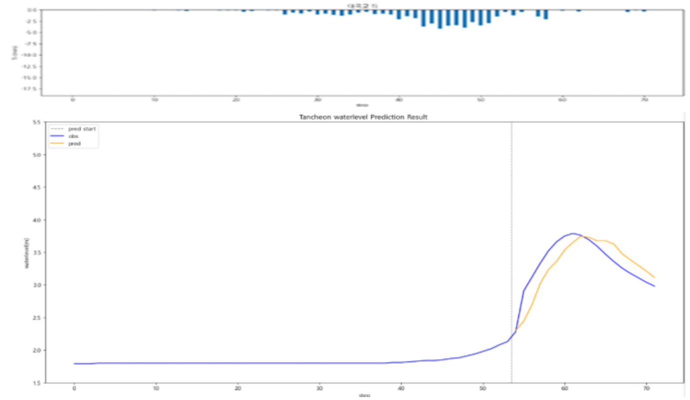
testdata_ 220712

R2 : -0.1308



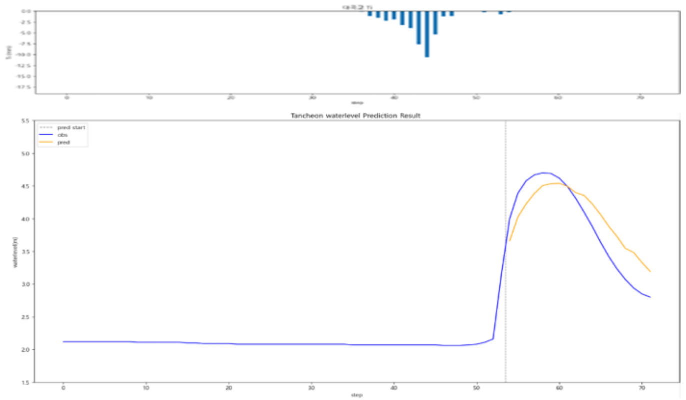
testdata_ 250716

R2 : 0.5949



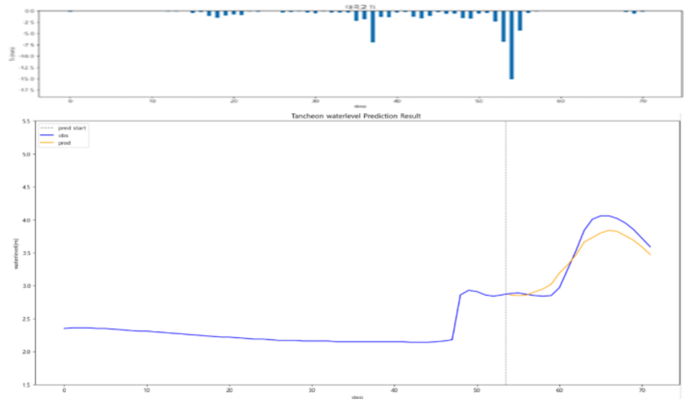
testdata_230711

R2 : 0.7288



testdata_250814

R2 : 0.8942



데이터 편집 (수위 X 0.98 , x 1.02)

학습데이터

- 관심 수위 : 궁내교 (2) 대곡교 (3.8)
- smoothing : 궁내교 (1) 대곡교 (2)
- 그래프 스케일 고정 : 궁내교 (0.7 - 3) 대곡교 (1.5 – 5.5)

- learning_rate = 0.0005
- Loss_function = MAE
- Seq_length = 36 (6시간)
- Num_of_layer = 18

홍수통제소 API 호출 → MIET 활용 강우 이벤트 추출 → 관심 수위 이상 추출

time	서울시 (대곡교)	성남시 (성남북초교)	광주시 (남한산초교)	성남시 (대장동)	성남시 (구미초교)	성남시 (한국학중앙연구원)	성남시 (궁내교)_WL	성남시 (궁내교)_Q	서울시 (대곡교)_WL	서울시 (대곡교)_Q	대곡교_Ti	궁내교_Ti
2014-07-24 15:30:00	0.0	0.0	0.0	0.0	0.0	0.0	1.12	5.11	2.17	40.56	0.0	0.0

학습 , 검증 : 총 67개 中 61개 데이터 파일 사용

테스트 : 학습 데이터에서 제외한 6개 데이터 파일 (4개 이벤트)

\+ 학습에서 검증에 사용된 2개 이벤트 사용

- 학습 : 2014 ~ 20230629 (43개)
- 검증 : 20230713 ~ 250813 (12개)
- 테스트 : 2550916 ~ 251006 (6개)

데이터 편집 (수위 X 0.98)

- 편향 보정 효과는 있음
- R^2 기준으로는 손해

기본 모델 대비 평균 R2 : 궁내교 ▼ 저하 (- 6.2%) , 대곡교 ▼ 저하 (- 1.5%)

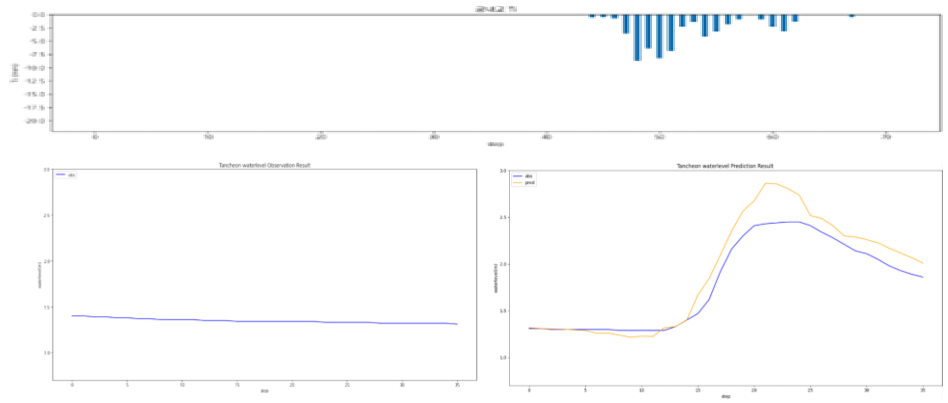
	궁내교				대곡교			
Test data	r2	mse	mae	rmse	r2	mse	mae	rmse
20.08.02	0.8492	0.8492	0.1350	0.1785	0.8579	0.0867	0.1858	0.2944
22.07.12	0.2864	0.0283	0.1397	0.1682	0.7718	0.0319	0.1560	0.1787
23.07.11	0.8814	0.0331	0.1257	0.1821	0.8246	0.1596	0.3170	0.3995
24.07.23	0.8244	0.0142	0.0692	0.1191	0.5221	0.0227	0.1072	0.1507
25.07.16	0.7627	0.0238	0.1369	0.1542	0.9736	0.0121	0.0856	0.1103
25.08.14	0.4144	0.0412	0.1441	0.2031	0.7718	0.0791	0.2257	0.2813

궁내교

* model_(gn/dg)_0.98

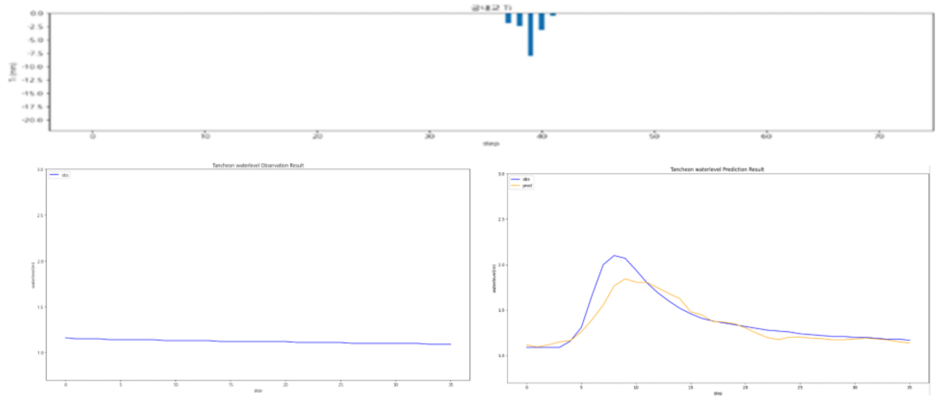
testdata_ 200802

R2 : 0.8492



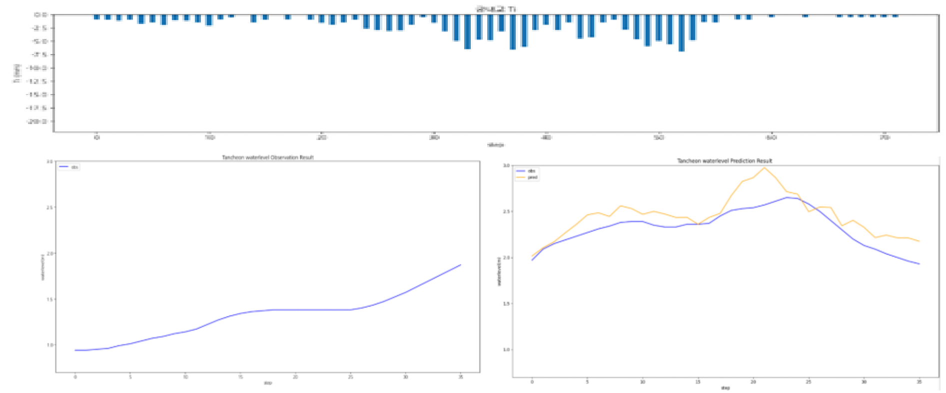
testdata_ 240723

R2 : 0.8244



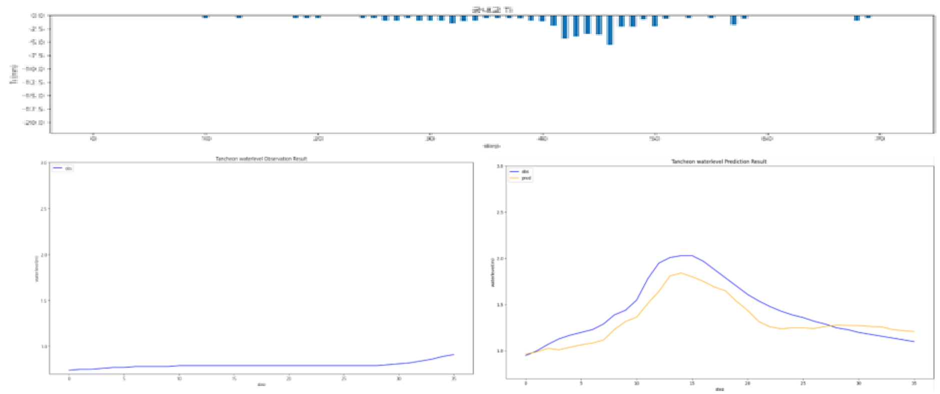
testdata_ 220712

R2 : 0.2864



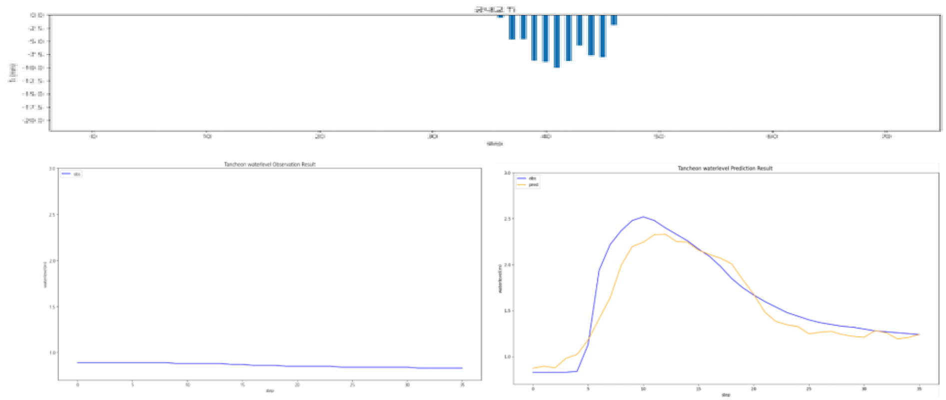
testdata_ 250716

R2 : 0.7627



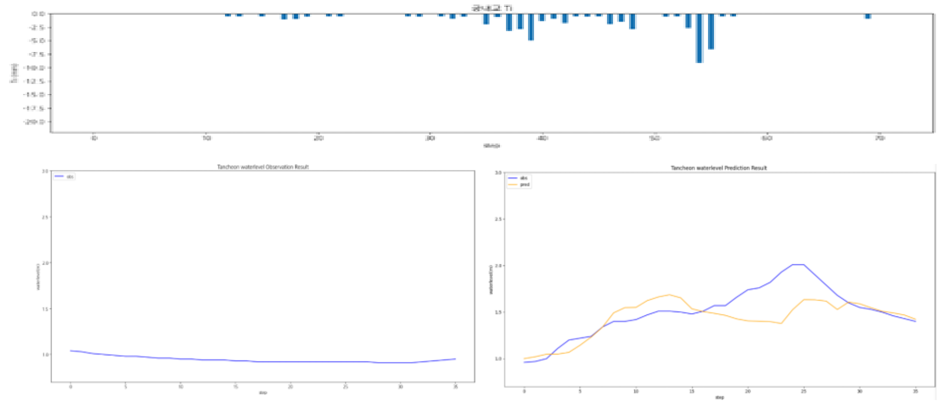
testdata_230711

R2 : 0.8814



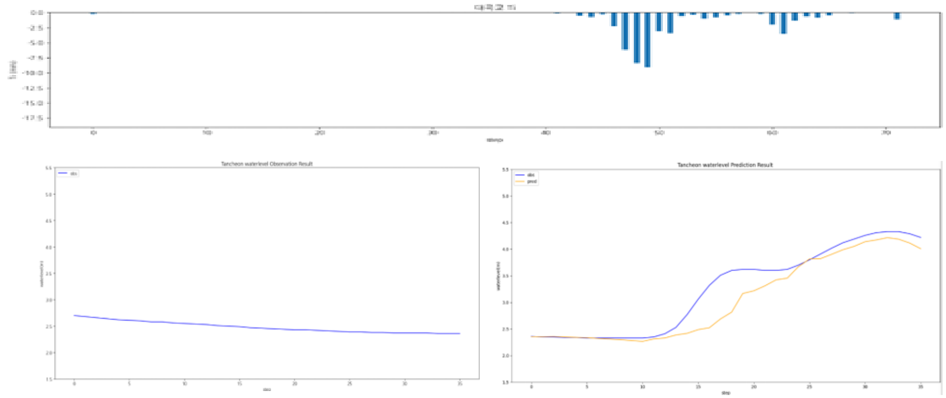
testdata_250814

R2 : 0.4144



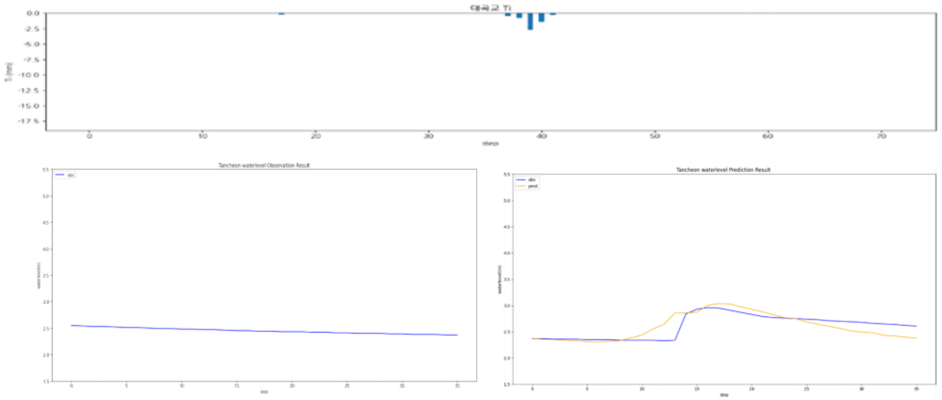
대곡교

* model_(gn/dg)_0.98



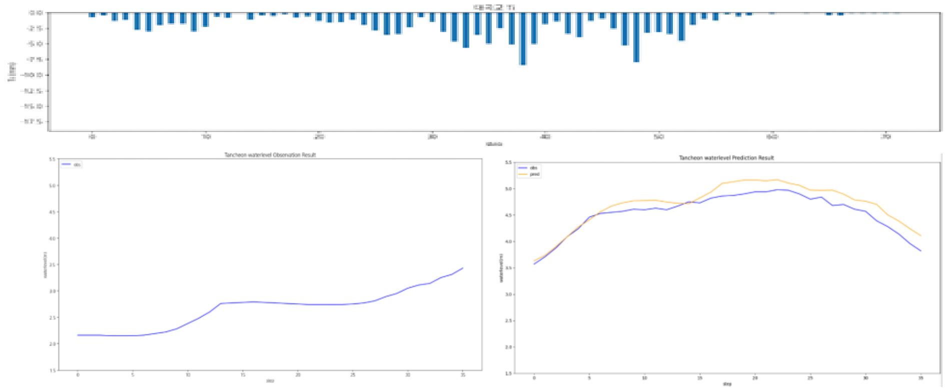
testdata_ 200802

R2 : 0.8579



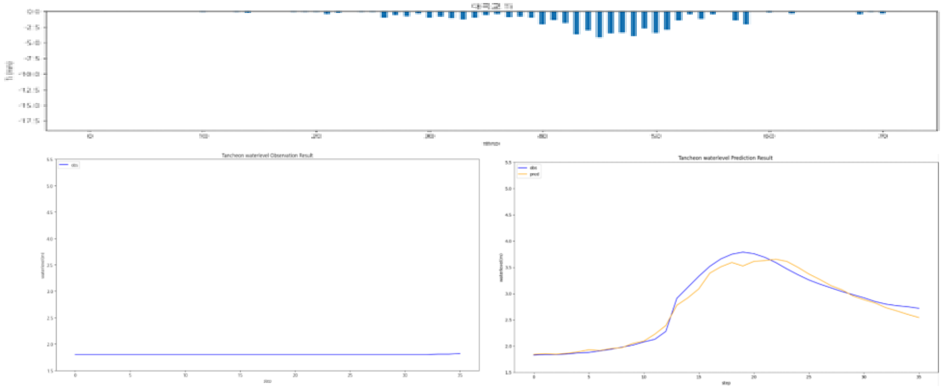
testdata_ 240723

R2 : 0.5221



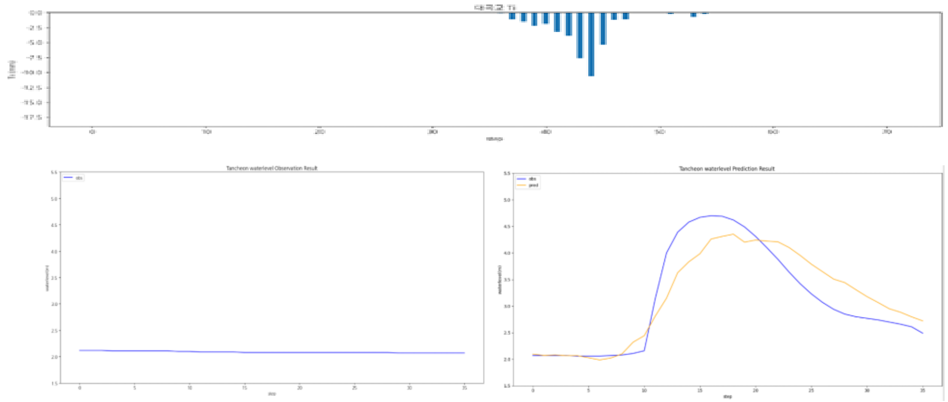
testdata_ 220712

R2 : 0.7718



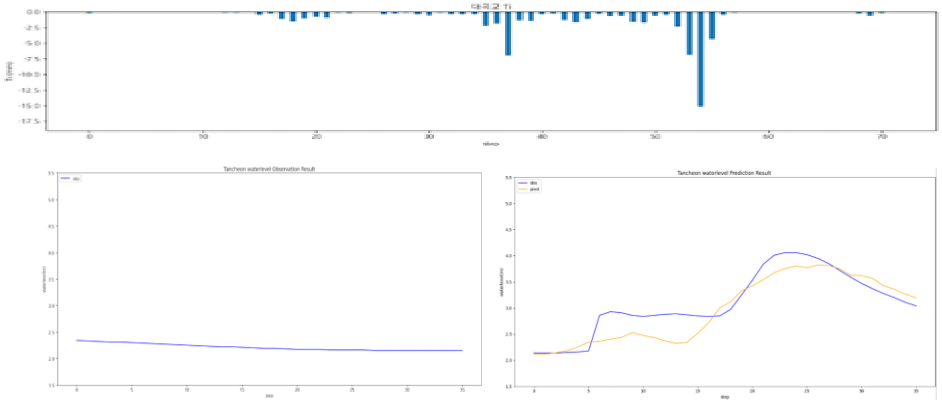
testdata_ 250716

R2 : 0.9736



testdata_230711

R2 : 0.8246



testdata_250814

R2 : 0.7718

데이터 편집 (수위 X 1.02)

- 과대보정 → 분산 증가
- peak 근처에서 불안정

기본 모델 대비 평균 R2 : 궁내교 ▼ 저하 (- 11.1%) , 대곡교 ▼저하 (- 6.1%)

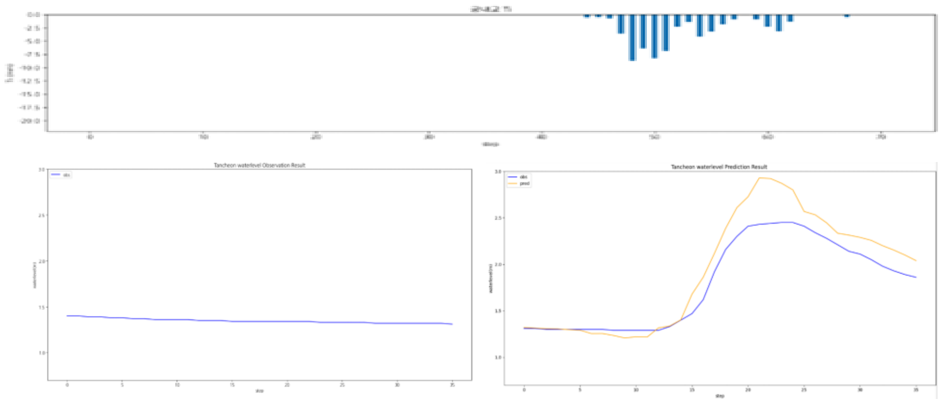
	궁내교				대곡교			
Test data	r2	mse	mae	rmse	r2	mse	mae	rmse
20.08.02	0.7922	0.0439	0.1592	0.2096	0.8688	0.0800	0.1656	0.2829
22.07.12	0.0156	0.0390	0.1680	0.1976	0.5974	0.0563	0.2138	0.2374
23.07.11	0.8994	0.0281	0.1186	0.1677	0.8086	0.1741	0.3343	0.4173
24.07.23	0.8400	0.0129	0.0691	0.1137	0.4905	0.0242	0.1086	0.1556
25.07.16	0.8186	0.0181	0.1220	0.1348	0.9726	0.0126	0.0917	0.1124
25.08.14	0.4433	0.0392	0.1474	0.1981	0.7639	0.0819	0.2331	0.2862

궁내교

* model_(gn/dg)_1.02

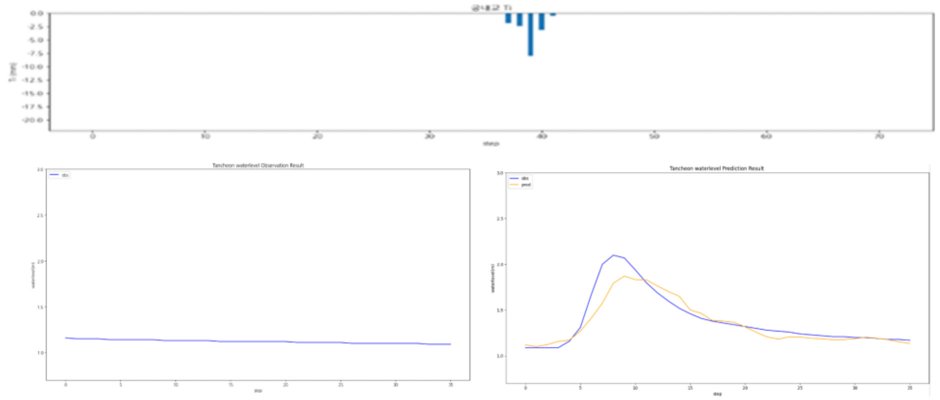
testdata_ 200802

R2 : 0.7922



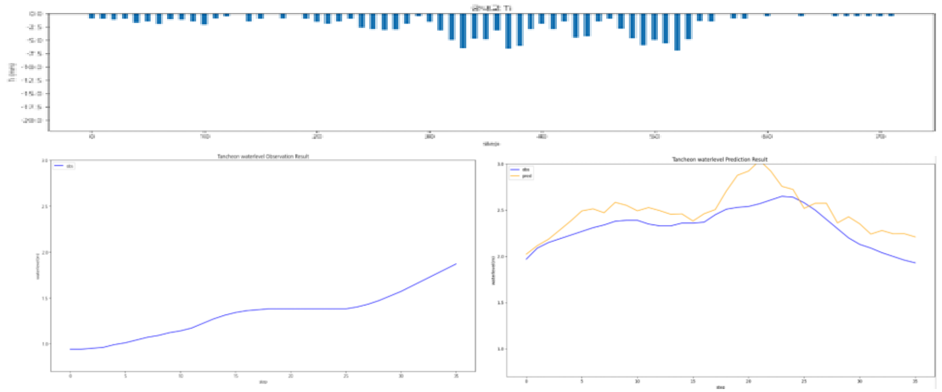
testdata_ 240723

R2 : 0.8400



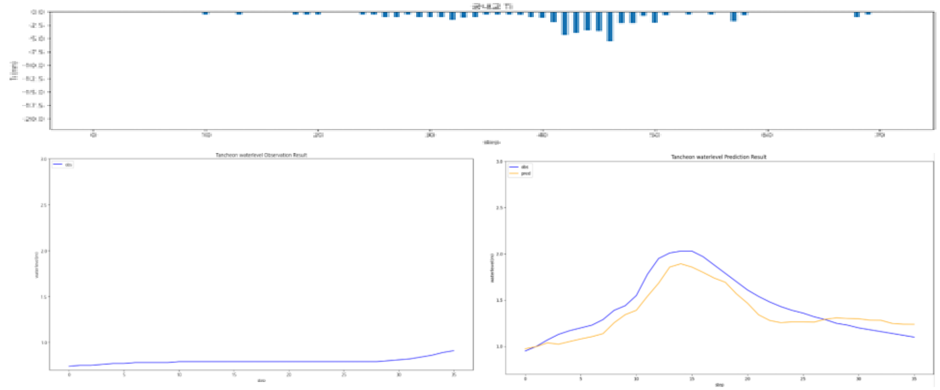
testdata_ 220712

R2 : 0.0156



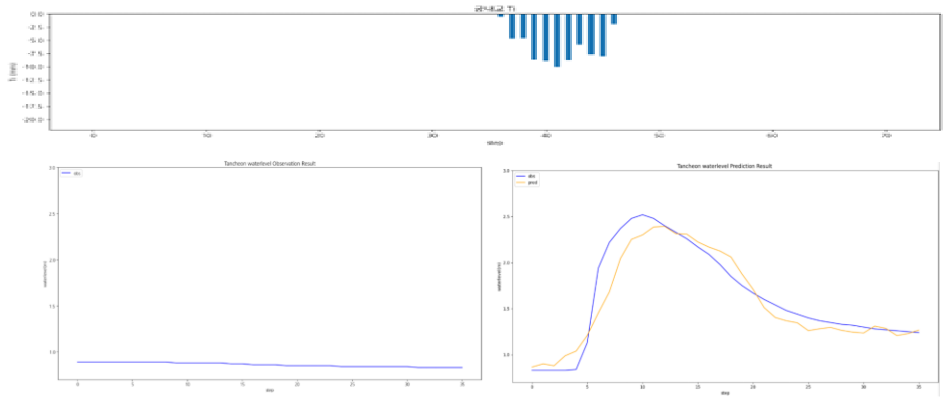
testdata_ 250716

R2 : 0.8186



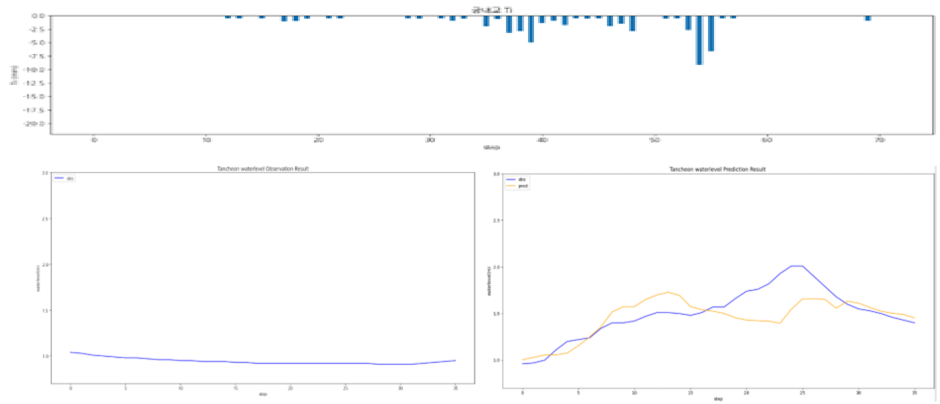
testdata_230711

R2 : 0.8994



testdata_250814

R2 : 0.4433

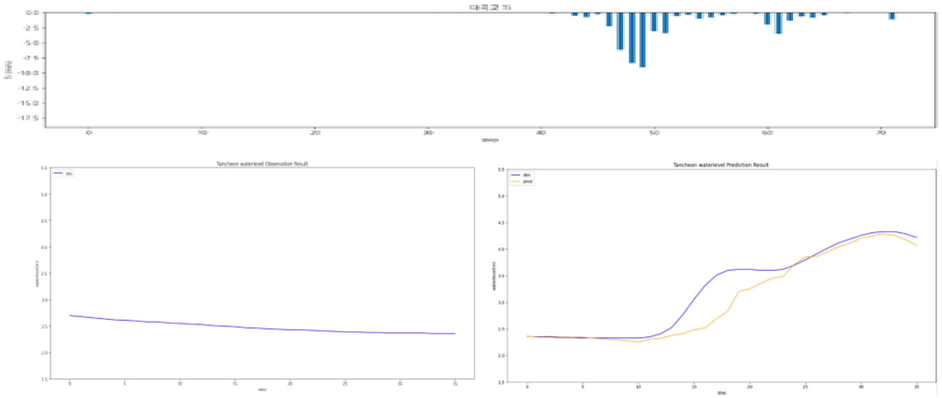


대곡교

* model_(gn/dg)_1.02

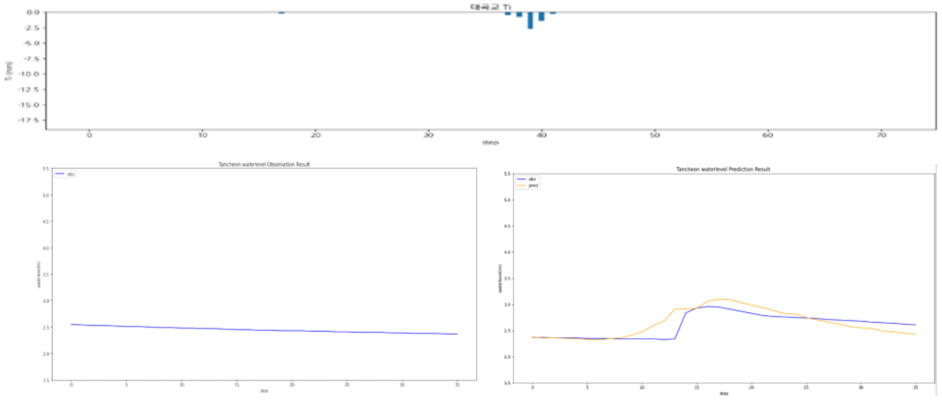
testdata_ 200802

R2 : 0.8688



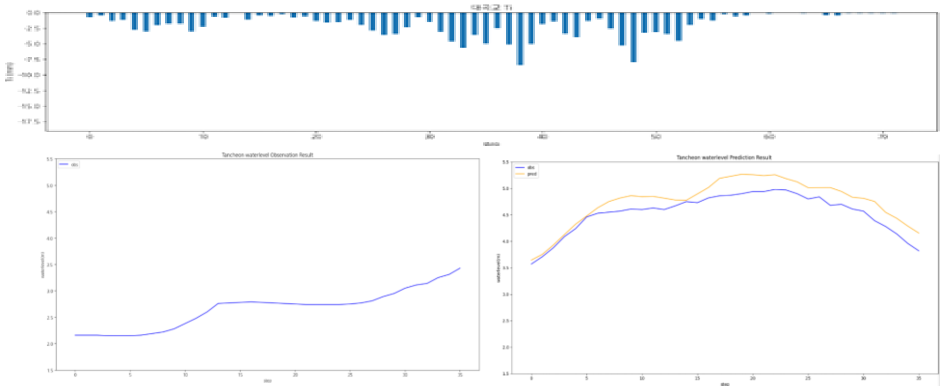
testdata_ 240723

R2 : 0.4905



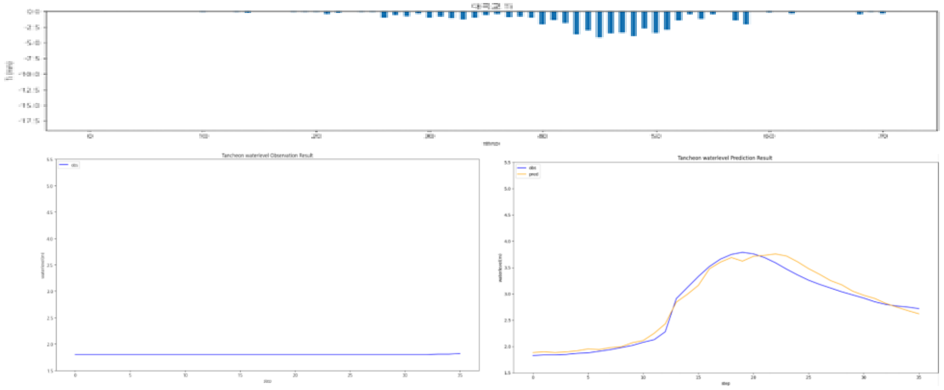
testdata_ 220712

R2 : 0.5974



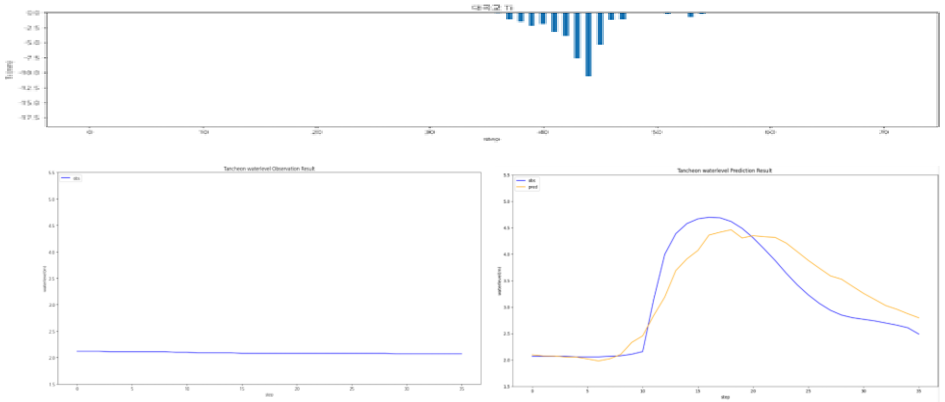
testdata_ 250716

R2 : 0.9726



testdata_230711

R2 : 0.8086



testdata_250814

R2 : 0.7639